7.10 Appendices

Appendix 7.A: The Okumura–Hata Model

Two forms of the Okumura-Hata model are available. In the first form, the pathloss (in dB) is written as

$$PL = PL_{\mathrm{freespace}} + A_{\mathrm{exc}} - H_{\mathrm{cb}} - H_{\mathrm{cm}} \tag{7.36}$$

where $PL_{\mathrm{freespace}}$ is the freespace pathloss, A_{exc} is the excess pathloss (as a function of distance and frequency) for a BS height $h_{\mathrm{b}} = 200$ m, and a UE height $h_{\mathrm{m}} = 3$ m; it is shown in Fig. 7.20. The correction factors H_{cb} and H_{cm} are shown in Figs. 7.21 and 7.22, respectively.

The more common form is a curve fitting of Okumura's original results.[9] In that implementation, the pathloss is written as

$$PL = A + B\log_{10}(d) + C$$

where A, B, and C are factors that depend on frequency and antenna height.

$$\begin{aligned} A &= 69.55 + 26.16\log_{10}(f_{\mathrm{c}}) - 13.82\log_{10}(h_{\mathrm{b}}) - a(h_{\mathrm{m}}) && (7.37)\\ B &= 44.9 - 6.55\log_{10}(h_{\mathrm{b}}) && (7.38) \end{aligned}$$

where f_{c} is given in MHz and d in km.

The function $a(h_{\mathrm{m}})$ and the factor C depend on the environment:

- small and medium-size cities:

$$\begin{aligned} a(h_{\mathrm{m}}) &= (1.1\log_{10}(f_{\mathrm{c}}) - 0.7)h_{\mathrm{m}} - (1.56\log_{10}(f_{\mathrm{c}}) - 0.8) && (7.39)\\ C &= 0\ . && (7.40) \end{aligned}$$

- metropolitan areas

$$\begin{aligned} a(h_{\mathrm{m}}) &= \begin{cases} 8.29(\log_{10}(1.54h_{\mathrm{m}})^2 - 1.1 & \text{for } f \leq 200 \text{ MHz}\\ 3.2(\log_{10}(11.75h_{\mathrm{m}})^2 - 4.97 & \text{for } f \geq 400 \text{ MHz} \end{cases} && (7.41)\\ C &= 0\ . && (7.42) \end{aligned}$$

- suburban environments

$$C = -2[\log_{10}(f_{\mathrm{c}}/28)]^2 - 5.4\ . \tag{7.43}$$

- rural area

$$C = -4.78[\log_{10}(f_{\mathrm{c}})]^2 + 18.33\log_{10}(f_{\mathrm{c}}) - 40.98\ . \tag{7.44}$$

  The function $a(h_{\mathrm{m}})$ in suburban and rural areas is the same as for urban (small and medium-sized cities) areas.

[9] Note that the results from the curve fitting can be slighltly different from the results of Eq. (7.36).

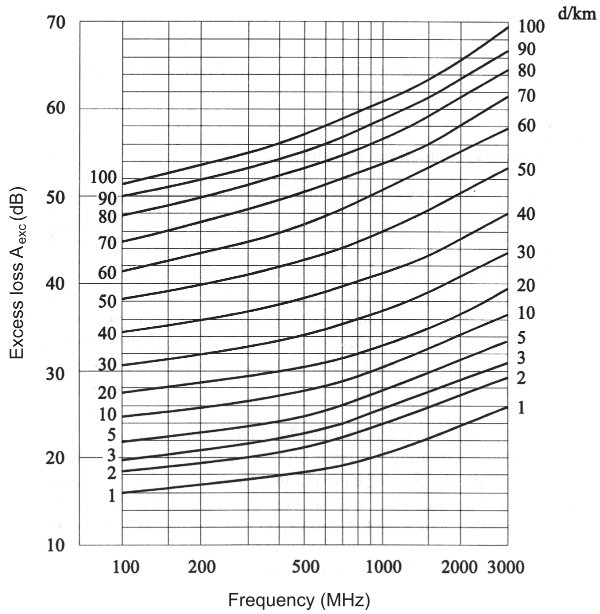

Figure 7.20: Pathloss according to the Okumura-Hata model. Excess loss is the difference between Okumura-Hata pathloss (at the reference points $h_b = 200m$, $h_m = 3m$), and free space loss. Reproduced from Okumura et al. [1968].

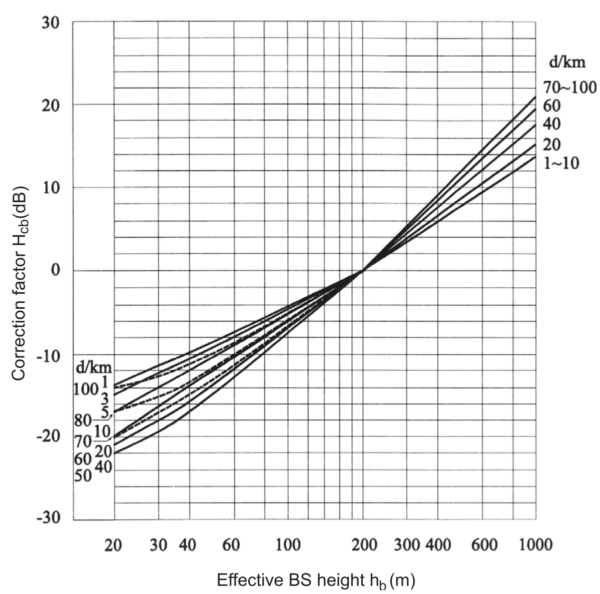

Figure 7.21: Correction factor for base station height. Reproduced from Okumura et al. [1968].

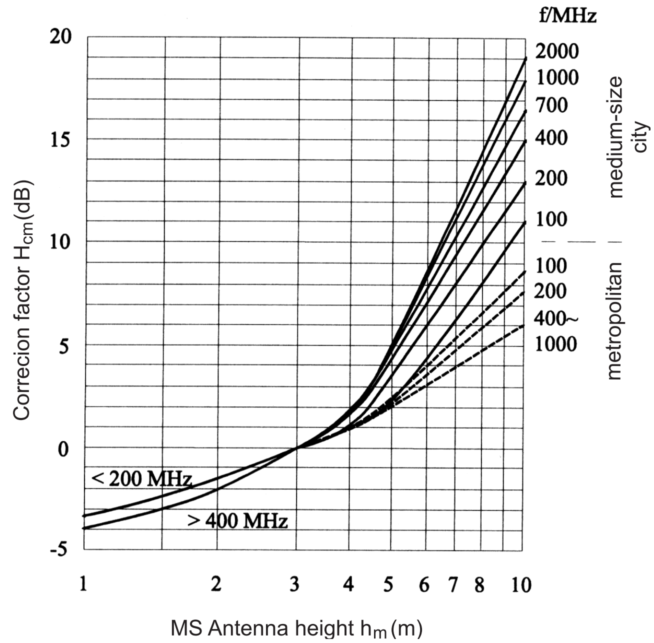

Figure 7.22: Correction factor for UE height. Reproduced from Okumura et al. [1968].

Table 7.2: Range of validity for the Okumura-Hata model.

carrier frequency	f_{c}	150...1500 MHz
effective BS-antenna height	h_{b}	30...200 m
effective UE-antenna height	h_{m}	1...10 m
distance	d	1...20 km

Table 7.2 gives the parameter range in which the model is valid. It is noteworthy that the parameter range does not encompass the 1800 MHz frequency range most commonly used for second-and third generation cellular systems. This problem was solved by the COST 231 - Hata model, which extends the validity region to the 1500 - 2000 MHz range by defining

$$\begin{aligned} A &= 46.3 + 33.9\log_{10}(f_{\mathrm{c}}) - 13.82\log_{10}(h_{\mathrm{b}}) - a(h_{\mathrm{m}}) \\ B &= 44.9 - 6.55\log_{10}(h_{\mathrm{b}}) \end{aligned} \tag{7.45}$$

where $a(h_m)$ is defined in Eq. (7.40). C is 0 in small and medium-sized cities, and 3 in metropolitan areas.

The Okumura-Hata model also assumes that there are no dominant obstacles between the BS and the UE, and that the terrain profile changes only slowly. [Badsberg et al. 1995] and [Isidoro et al. 1995] suggested correction factors that are intended to avoid these restrictions; however, they have not found the widespread acceptance of the COST 231 correction factors.

Appendix 7.B: The COST 231–Walfish–Ikegami Model

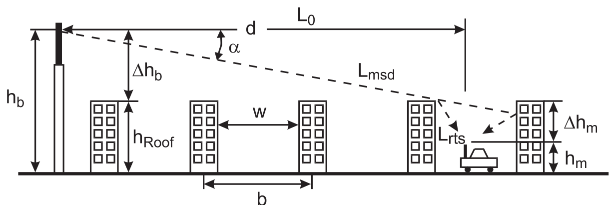

Figure 7.23: Parameters in the COST 231-Walfish-Ikegami model. Reproduced with permission from Damosso and Correia [1999] © European Union.

The COST 231 model is a pathloss model for the case of small distances between UE and BS, and/or small height of the UE. The total pathloss for the LOS case is given by

$$PL = 42.6 + 26\log_{10}(d) + 20\log_{10}(f_c) \tag{7.46}$$

for $d \geq 20$ m, where again d is in units of kilometers, and f_c is in units of MHz.

For the NLOS case, the pathloss consists of the free-space pathloss L_0, the *multiscreen loss* L_{msd} along the propagation path, and the attenuation from the last roofedge to the UE, L_{rts} (*roof-top-to-street diffraction and scatter loss*):

$$PL = \begin{cases} PL_0 + L_{rts} + L_{msd} & \text{for } L_{rts} + L_{msd} > 0 \\ PL_0 & \text{for } L_{rts} + L_{msd} \leq 0 \end{cases} . \tag{7.47}$$

The free-space pathloss is

$$L_0 = 32.4 + 20\log_{10} d + 20\log_{10} f_c \ . \tag{7.48}$$

Ikegami derived the diffraction loss L_{rts} as

$$L_{rts} = -16.9 - 10\log_{10} w + 10\log_{10} f_c + 20\log_{10} \Delta h_m + L_{ori} \tag{7.49}$$

where w is the width of the street in meters, and

$$\Delta h_m = h_{Roof} - h_m \tag{7.50}$$

is the difference between the building height h_{Roof} and the height of the UE h_m. The orientation of the street is taken into account by an empirical correction factor L_{ori}

$$L_{ori} = \begin{cases} -10 + 0.354\varphi & \text{for } 0° \leq \varphi \leq 35° \\ 2.5 + 0.075(\varphi - 35) & \text{for } 35° \leq \varphi \leq 55° \\ 4.0 - 0.114(\varphi - 55) & \text{for } 55° \leq \varphi \leq 90° \end{cases} \tag{7.51}$$

where φ is the angle between the street orientation and the direction of incidence in degrees.

For the computation of the multiscreen loss L_{msd}, building edges are modeled as screens. The multiscreen loss is then given as [Walfish and Bertoni 1988]:

$$L_{msd} = L_{bsh} + k_a + k_d \log_{10} d + k_f \log_{10} f_c - 9\log_{10} b \tag{7.52}$$

where b is the distance between two buildings (in meters). Furthermore,

$$L_{bsh} = \begin{cases} -18\log_{10}(1 + \Delta h_b) & \text{for } h_b > h_{Roof} \\ 0 & \text{for } h_b \leq h_{Roof} \end{cases} \tag{7.53}$$

$$k_a = \begin{cases} 54 & \text{for } h_b > h_{Roof} \\ 54 - 0.8\Delta h_b & \text{for } d \geq 0.5 \text{ km and } h_b \leq h_{Roof} \\ 54 - 0.8\Delta h_b \dfrac{d}{0.5} & \text{for } d < 0.5 \text{ km and } h_b \leq h_{Roof} \end{cases} \tag{7.54}$$

where

$$\Delta h_{\mathrm{b}} = h_{\mathrm{b}} - h_{\mathrm{Roof}} \tag{7.55}$$

and h_{b} is the height of the BS. The dependence of the pathloss on the frequency and distance is given via the parameters k_{d} and k_{f} in Eq. (7.52):

$$k_{\mathrm{d}} = \begin{cases} 18 & \text{for } h_{\mathrm{b}} > h_{\mathrm{Roof}} \\ 18 - 15\dfrac{\Delta h_{\mathrm{b}}}{h_{\mathrm{Roof}}} & \text{for } h_{\mathrm{b}} \leq h_{\mathrm{Roof}} \end{cases} \tag{7.56}$$

$$k_{\mathrm{f}} = -4 + \begin{cases} 0.7\left(\dfrac{f_{\mathrm{c}}}{925} - 1\right) & \begin{array}{l}\text{for medium-size cities} \\ \text{suburban areas with average vegetation density}\end{array} \\ 1.5\left(\dfrac{f_c}{925} - 1\right) & \text{for metropolitan areas} \end{cases} . \tag{7.57}$$

Table 7.3 gives the validity range for this model

Further assumptions include a Manhatten grid (streets intersecting at right angles), constant building height, and flat terrain. Furthermore, the model does not include the effect of waveguiding through street canyons, which can lead to an underestimation of the received fieldstrength.

carrier frequency	f_{c}	800...2000 MHz
height of the BS antenna	h_{b}	4...50 m
height of the UE antenna	h_{m}	1...3 m
distance	d	0.02...5 km

Table 7.3: Range of validity for the COST 231-Walfish-Ikegami model.

Appendix 7.C: The COST 207 GSM Model

The COST 207 model [Failii 1989] gives normalized scattering functions, as well as amplitude statistics, for four classes of environments: Rural Area, Typical Urban, Bad Urban, and Hilly Terrain. The following PDPs are used:

- Rural Area (RA)

$$P_h(\tau) = \begin{cases} \exp\left(-9.2\frac{\tau}{\mu\mathrm{s}}\right) & \text{for } 0 < \tau < 0.7\mu\mathrm{s} \\ 0 & \text{elsewhere} \end{cases} . \tag{7.58}$$

- Typical Urban area (TU)

$$P_h(\tau) = \begin{cases} \exp\left(-\frac{\tau}{\mu\mathrm{s}}\right) & \text{for } 0 < \tau < 7\mu\mathrm{s} \\ 0 & \text{elsewhere} \end{cases} . \tag{7.59}$$

- Bad Urban area (BU)

$$P_h(\tau) = \begin{cases} \exp\left(-\frac{\tau}{\mu\mathrm{s}}\right) & \text{for } 0 < \tau < 5\mu\mathrm{s} \\ 0.5\exp\left(5-\frac{\tau}{\mu\mathrm{s}}\right) & \text{for } 5 < \tau < 10\mu\mathrm{s} \\ 0 & \text{elsewhere} \end{cases} . \tag{7.60}$$

- Hilly Terrain (HT)

$$P_h(\tau) = \begin{cases} \exp\left(-3.5\frac{\tau}{\mu\mathrm{s}}\right) & \text{for } 0 < \tau < 2\mu\mathrm{s} \\ 0.1\exp\left(15-\frac{\tau}{\mu\mathrm{s}}\right) & \text{for } 15 < \tau < 20\mu\mathrm{s} \\ 0 & \text{elsewhere} \end{cases} . \tag{7.61}$$

In a tapped delay-line realization, each channel has a Doppler spectrum $P_s(\nu, \tau_i)$, where i is the index of the tap. The following four types of Doppler spectra are defined in the COST 207 model, where $\nu_{\max}$ represents the maximum Doppler shift and $G(A, \nu_1, \nu_2)$ is the Gaussian function

$$G(A, \nu_1, \nu_2) = A\exp\left(-\frac{(\nu-\nu_1)^2}{2\nu_2^2}\right) \tag{7.62}$$

and A is a normalization constant chosen so that $\int P_s(\nu, \tau_i)d\nu = 1$.

a) *CLASS* is the classical (Jakes) Doppler spectrum and is used for paths with delays not in excess of 500 ns ($\tau_i \le 0.5$ μs):

$$P_s(\nu, \tau_i) = \frac{A}{\sqrt{1-\left(\frac{\nu}{\nu_{\max}}\right)^2}} \tag{7.63}$$

for $\nu \in [-\nu_{\max}, \nu_{\max}]$;

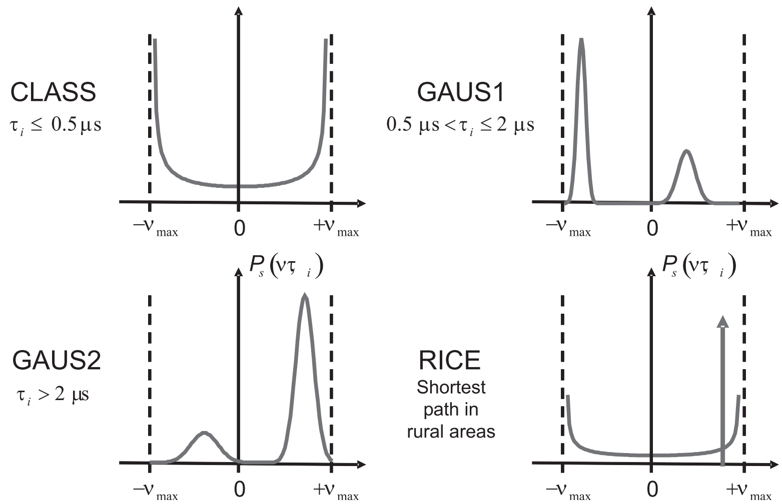

Figure 7.24: Doppler spectra used in GSM channel models.

b) *GAUS1* is the sum of two Gaussian functions and is used for excess delay times in the range of 500 ns to 2 μs (0.5 $\mu s \leq \tau_i \leq 2$ μs):

$$P_s(\nu, \tau_i) = G(A, -0.8\nu_{\max}, 0.05\nu_{\max}) + G(A_1, 0.4\nu_{\max}, 0.1\nu_{\max}) \tag{7.64}$$

where A_1 is 10 dB smaller than A;

c) *GAUS2* is also the sum of two Gaussian functions and is used for paths with delays in excess of 2 μs ($\tau_i \geq 2$ μs):

$$P_s(\nu, \tau_i) = G(B, 0.7\nu_{\max}, 0.1\nu_{\max}) + G(B_1, -0.4\nu_{\max}, 0.15\nu_{\max}) \tag{7.65}$$

where B_1 is 15 dB smaller than B;

d) *RICE* is the sum of a classical Doppler spectrum and one direct path, such that the total multipath contribution is equal to that of the direct path. This spectrum is used for the shortest path of the model for propagation in rural areas:

$$P_s(\nu, \tau_i) = \frac{0.41}{2\pi\nu_{\max}\sqrt{1 - \left(\frac{\nu}{\nu_{\max}}\right)^2}} + 0.91\delta(\nu - 0.7\nu_{\max}) \tag{7.66}$$

for $\nu \in [-\nu_{\max}, \nu_{\max}]$;

Tables 7.4-7.7 give tapped delay line implementations. Note that the COST 207 final report provides a variety of tap settings that all approximate the continuous profiles; the settings shown here are just samples with comparatively small number of taps.

Tap#	Delay [μs]	Power [dB]	Doppler category
1	0	0	RICE
2	0.2	-2	CLASS
3	0.4	-10	CLASS
4	0.6	-20	CLASS

Table 7.4: Parameters for rural (non-hilly) area (RA)

Tap#	Delay [μs]	Power [dB]	Doppler category
1	0	-3	CLASS
2	0.2	0	CLASS
3	0.6	-2	GAUS1
4	1.6	-6	GAUS1
5	2.4	-8	GAUS2
6	5.0	-10	GAUS2

Table 7.5: Parameters for urban (non-hilly) area (TU)

Tap#	Delay [μs]	Power [dB]	Doppler category
1	0	-3	CLASS
2	0.4	0	CLASS
3	1.0	-3	GAUS1
4	1.6	-5	GAUS1
5	5.0	-2	GAUS2
6	6.6	-4	GAUS2

Table 7.6: Parameters for hilly urban area (BU)

Tap#	Delay [μs]	Power [dB]	Doppler category
1	0	0	CLASS
2	0.2	-2	CLASS
3	0.4	-4	CLASS
4	0.6	-7	CLASS
5	15	-6	GAUS2
6	17.2	-12	GAUS2

Table 7.7: Parameters for hilly terrain (HT)

Appendix 7.D: The 3GPP Spatial Channel Model

In 2001, the 3GPP and 3GPP2 industry alliances decided to jointly develop channel models that are to be used for the evaluation of cellular systems with multiple antenna elements, which was published in 2004 (a peer-reviewed journal paper based on this model is [Calcev et al. 2007]).The original model was quite restricted in terms of the bandwidth (5 MHz) as well as environments. Generalizations were developed based on the measurements of the European Winner and Winner II models, and were used for the evaluation of 4G systems. Further work was done by 3GPP for extension to 5G, resulting in a model that was published as TR 38.901, claiming to have validity up to 100 GHz carrier frequency. This Appendix describes this latest (at the time of the writing of the 3rd edition of this book) iteration of this model.

Environments

The model is defined for the following environments

- **Urban Macrocell, UMa.** This is a scenario with dense buildup. BS height is above the surrounding rooftops, and its "typical" (i.e., recommended value) is 25m, with typical minimum distance between BS and UE 35 m, and typical inter-site distance of 500m. The UE can be located either outdoor (assumed to occur 20% of the time) or indoor (80%).
- **Urban Microcell, UMi.** This scenario is very similar to UMa, but assumes BS height below the surrounding rooftops, and typically at 10 m height above ground. Typical minimum BS-UE distance is 10m, and typical inter-site distance is 200 m. Outdoor/indoor parameters of the UE are the same as for UMa.
- **Rural Macrocell, RMa.** This scenario is only defined for up to $7GHz$ carrier frequency. The buildup of these environments is much sparser, and only 50% of users are assumed to be indoor. Typical BS height is 35 m.
- **Indoor, In.** The indoor environment has a number of sub-environments, reflecting the different types of buildings. The most frequently used is an indoor office, with the BS typically at ceiling height (3 m), the UE at 1 m height, and typical room sizes 120×50 m. The indoor factory, InF, environment is divided into futher sub-categories, according to clutter density (sparse S or dense D), and BS height (low L or high H), giving rise to the InF-SL, InF-DL, InF-SH, and InF-DH environments. In addition, the InF-HH (high Tx, high Rx) is also defined, and can be seen as a sort of "factory backhaul" that always has LOS.

Pathloss

The basic structure of the 3GPP pathloss model follows the Okomura-Hata model, with the fitting parameters related to the geometry in the environment. The tables on the next pages provide the specifications for the cases defined in 38.901. They provide equations for both the LOS and the NLOS situations. The LOS probability has already been described in Sec. 7.1.7.

For outdoor-to-indoor propagation, the total pathloss is given as

$$\mathrm{PL} = \mathrm{PL}_{\mathrm{out}} + \mathrm{PL}_{\mathrm{tw}} + \mathrm{PL}_{\mathrm{in}} + N\left(0, \sigma_P^2\right) \tag{7.67}$$

where $\mathrm{PL}_{\mathrm{out}}$ is the outdoor pathloss (using the distance between the BS and the sum of outdoor and indoor distance), $\mathrm{L}_{\mathrm{out}}$ is the indoor pathloss, which is modeled as a function of $d_{\mathrm{2D-in}}$. The deterministic part of the penetration loss, $\mathrm{PL}_{\mathrm{tw}}$ is

$$\mathrm{PL}_{tw} = \mathrm{PL}_{npi} - 10\log_{10} \sum_{i=1}^{N} \left(p_i \times 10^{\frac{L_{material\_i}}{-10}}\right) \tag{7.68}$$

Scenario	LOS/NLOS	Pathloss [dB], f_c is in GHz and d is in meters, see note 6	Shadow fading std [dB]	Applicability range, antenna height default values
RMa	LOS	$PL_{\mathrm{RMa-LOS}} = \begin{cases} PL_1 & 10\mathrm{m} \le d_{\mathrm{2D}} \le d_{\mathrm{BP}} \\ PL_2 & d_{\mathrm{BP}} \le d_{\mathrm{2D}} \le 10\mathrm{km} \end{cases}$, see note 5 $PL_1 = 20\log_{10}(40\pi d_{\mathrm{3D}} f_c / 3) + \min(0.03h^{1.72}, 10)\log_{10}(d_{\mathrm{3D}}) - \min(0.044h^{1.72}, 14.77) + 0.002\log_{10}(h) d_{\mathrm{3D}}$ $PL_2 = PL_1(d_{\mathrm{BP}}) + 40\log_{10}(d_{\mathrm{3D}} / d_{\mathrm{BP}})$	$\sigma_{\mathrm{SF}} = 4$ $\sigma_{\mathrm{SF}} = 6$	$h_{\mathrm{BS}} = 35\mathrm{m}$ $h_{\mathrm{UT}} = 1.5\mathrm{m}$ $W = 20\mathrm{m}$ $h = 5\mathrm{m}$ h = avg. building height W = avg. street width The applicability ranges: $5\mathrm{m} \le h \le 50\mathrm{m}$ $5\mathrm{m} \le W \le 50\mathrm{m}$ $10\mathrm{m} \le h_{\mathrm{BS}} \le 150\mathrm{m}$ $1\mathrm{m} \le h_{\mathrm{UT}} \le 10\mathrm{m}$
	NLOS	$PL_{\mathrm{RMa-NLOS}} = \max(PL_{\mathrm{RMa-LOS}}, PL'_{\mathrm{RMa-NLOS}})$ for $10\mathrm{m} \le d_{\mathrm{2D}} \le 5\mathrm{km}$ $PL'_{\mathrm{RMa-NLOS}} = 161.04 - 7.1\log_{10}(W) + 7.5\log_{10}(h) - (24.37 - 3.7(h/h_{\mathrm{BS}})^2)\log_{10}(h_{\mathrm{BS}}) + (43.42 - 3.1\log_{10}(h_{\mathrm{BS}}))(\log_{10}(d_{\mathrm{3D}}) - 3) + 20\log_{10}(f_c) - (3.2(\log_{10}(11.75h_{\mathrm{UT}}))^2 - 4.97)$	$\sigma_{\mathrm{SF}} = 8$	
UMa	LOS	$PL_{\mathrm{UMa-LOS}} = \begin{cases} PL_1 & 10\mathrm{m} \le d_{\mathrm{2D}} \le d'_{\mathrm{BP}} \\ PL_2 & d'_{\mathrm{BP}} \le d_{\mathrm{2D}} \le 5\mathrm{km} \end{cases}$, see note 1 $PL_1 = 28.0 + 22\log_{10}(d_{\mathrm{3D}}) + 20\log_{10}(f_c)$ $PL_2 = 28.0 + 40\log_{10}(d_{\mathrm{3D}}) + 20\log_{10}(f_c) - 9\log_{10}((d'_{\mathrm{BP}})^2 + (h_{\mathrm{BS}} - h_{\mathrm{UT}})^2)$	$\sigma_{\mathrm{SF}} = 4$	$1.5\mathrm{m} \le h_{\mathrm{UT}} \le 22.5\mathrm{m}$ $h_{\mathrm{BS}} = 25\mathrm{m}$
	NLOS	$PL_{\mathrm{UMa-NLOS}} = \max(PL_{\mathrm{UMa-LOS}}, PL'_{\mathrm{UMa-NLOS}})$ for $10\mathrm{m} \le d_{\mathrm{2D}} \le 5\mathrm{km}$ $PL'_{\mathrm{UMa-NLOS}} = 13.54 + 39.08\log_{10}(d_{\mathrm{3D}}) + 20\log_{10}(f_c) - 0.6(h_{\mathrm{UT}} - 1.5)$	$\sigma_{\mathrm{SF}} = 6$	$1.5\mathrm{m} \le h_{\mathrm{UT}} \le 22.5\mathrm{m}$ $h_{\mathrm{BS}} = 25\mathrm{m}$ Explanations: see note 3
		Optional $\mathrm{PL} = 32.4 + 20\log_{10}(f_c) + 30\log_{10}(d_{\mathrm{3D}})$	$\sigma_{\mathrm{SF}} = 7.8$	
UMi - Street Canyon	LOS	$PL_{\mathrm{UMi-LOS}} = \begin{cases} PL_1 & 10\mathrm{m} \le d_{\mathrm{2D}} \le d'_{\mathrm{BP}} \\ PL_2 & d'_{\mathrm{BP}} \le d_{\mathrm{2D}} \le 5\mathrm{km} \end{cases}$, see note 1 $PL_1 = 32.4 + 21\log_{10}(d_{\mathrm{3D}}) + 20\log_{10}(f_c)$ $PL_2 = 32.4 + 40\log_{10}(d_{\mathrm{3D}}) + 20\log_{10}(f_c) - 9.5\log_{10}((d'_{\mathrm{BP}})^2 + (h_{\mathrm{BS}} - h_{\mathrm{UT}})^2)$	$\sigma_{\mathrm{SF}} = 4$	$1.5\mathrm{m} \le h_{\mathrm{UT}} \le 22.5\mathrm{m}$ $h_{\mathrm{BS}} = 10\mathrm{m}$
	NLOS	$PL_{\mathrm{UMi-NLOS}} = \max(PL_{\mathrm{UMi-LOS}}, PL'_{\mathrm{UMi-NLOS}})$ for $10\mathrm{m} \le d_{\mathrm{2D}} \le 5\mathrm{km}$ $PL'_{\mathrm{UMi-NLOS}} = 35.3\log_{10}(d_{\mathrm{3D}}) + 22.4 + 21.3\log_{10}(f_c) - 0.3(h_{\mathrm{UT}} - 1.5)$	$\sigma_{\mathrm{SF}} = 7.82$	$1.5\mathrm{m} \le h_{\mathrm{UT}} \le 22.5\mathrm{m}$ $h_{\mathrm{BS}} = 10\mathrm{m}$ Explanations: see note 4
		Optional $\mathrm{PL} = 32.4 + 20\log_{10}(f_c) + 31.9\log_{10}(d_{\mathrm{3D}})$	$\sigma_{\mathrm{SF}} = 8.2$	

Pathloss equations for the different environments in 3GPP. From [3GPP 38.901]. © 3GPP.

InH - Office	LOS	$PL_{\mathrm{InH-LOS}} = 32.4 + 17.3\log_{10}(d_{\mathrm{3D}}) + 20\log_{10}(f_c)$	$\sigma_{\mathrm{SF}} = 3$	$1\mathrm{m} \le d_{\mathrm{3D}} \le 150\mathrm{m}$
	NLOS	$PL_{\mathrm{InH-NLOS}} = \max(PL_{\mathrm{InH-LOS}}, PL'_{\mathrm{InH-NLOS}})$ $PL'_{\mathrm{InH-NLOS}} = 38.3\log_{10}(d_{\mathrm{3D}}) + 17.30 + 24.9\log_{10}(f_c)$	$\sigma_{\mathrm{SF}} = 8.03$	$1\mathrm{m} \le d_{\mathrm{3D}} \le 150\mathrm{m}$
		Optional $PL'_{\mathrm{InH-NLOS}} = 32.4 + 20\log_{10}(f_c) + 31.9\log_{10}(d_{\mathrm{3D}})$	$\sigma_{\mathrm{SF}} = 8.29$	$1\mathrm{m} \le d_{\mathrm{3D}} \le 150\mathrm{m}$
InF	LOS	$PL_{LOS} = 31.84 + 21.50\log_{10}(d_{3D}) + 19.00\log_{10}(f_c)$	$\sigma_{SF} = 4.$	$1 \le d_{3D} \le 600\ m$
	NLOS	InF-SL: $PL = 33 + 25.5\log_{10}(d_{3D}) + 20\log_{10}(f_c)$ $PL_{NLOS} = \max(PL, PL_{LOS})$	$\sigma_{SF} = 5.7$	
		InF-DL: $PL = 18.6 + 35.7\log_{10}(d_{3D}) + 20\log_{10}(f_c)$ $PL_{NLOS} = \max(PL, PL_{LOS}, PL_{InF-SL})$	$\sigma_{SF} = 7.2$	
		InF-SH: $PL = 32.4 + 23.0\log_{10}(d_{3D}) + 20\log_{10}(f_c)$ $PL_{NLOS} = \max(PL, PL_{LOS})$	$\sigma_{SF} = 5.9$	
		InF-DH: $PL = 33.63 + 21.9\log_{10}(d_{3D}) + 20\log_{10}(f_c)$ $PL_{NLOS} = \max(PL, PL_{LOS})$	$\sigma_{SF} = 4.0$	

Note 1: Breakpoint distance $d'_{\mathrm{BP}} = 4\, h'_{\mathrm{BS}}\, h'_{\mathrm{UT}}\, f_c/c$, where f_c is the centre frequency in Hz, $c = 3.0\times10^8$ m/s is the propagation velocity in free space, and h'_{BS} and h'_{UT} are the effective antenna heights at the BS and the UT, respectively. The effective antenna heights h'_{BS} and h'_{UT} are computed as follows: $h'_{\mathrm{BS}} = h_{\mathrm{BS}} - h_{\mathrm{E}}$, $h'_{\mathrm{UT}} = h_{\mathrm{UT}} - h_{\mathrm{E}}$, where h_{BS} and h_{UT} are the actual antenna heights, and h_{E} is the effective environment height. For UMi $h_{\mathrm{E}} = 1.0$m. For UMa h_{E}=1m with a probability equal to $1/(1+C(d_{2D}, h_{UT}))$ and chosen from a discrete uniform distribution uniform(12,15,…,(h_{UT}-1.5)) otherwise. With $C(d_{2D}, h_{UT})$ given by

$$C(d_{\mathrm{2D}}, h_{\mathrm{UT}}) = \begin{cases} 0 & , h_{\mathrm{UT}} < 13\mathrm{m} \\ \left(\dfrac{h_{\mathrm{UT}} - 13}{10}\right)^{1.5} g(d_{\mathrm{2D}}) & , 13\mathrm{m} \le h_{\mathrm{UT}} \le 23\mathrm{m} \end{cases},$$

where

$$g(d_{\mathrm{2D}}) = \begin{cases} 0 & , d_{\mathrm{2D}} \le 18\mathrm{m} \\ \dfrac{5}{4}\left(\dfrac{d_{\mathrm{2D}}}{100}\right)^3 \exp\left(\dfrac{-d_{\mathrm{2D}}}{150}\right) & , 18\mathrm{m} < d_{\mathrm{2D}} \end{cases}.$$

Note that h_{E} depends on d_{2D} and h_{UT} and thus needs to be independently determined for every link between BS sites and UTs. A BS site may be a single BS or multiple co-located BSs.

Note 2: The applicable frequency range of the PL formula in this table is $0.5 < f_c < f_{\mathrm{H}}$ GHz, where $f_{\mathrm{H}} = 30$ GHz for RMa and $f_{\mathrm{H}} = 100$ GHz for all the other scenarios. It is noted that RMa pathloss model for >7 GHz is validated based on a single measurement campaign conducted at 24 GHz.

Note 3: UMa NLOS pathloss is from TR36.873 with simplified format and $PL_{\mathrm{UMa-LOS}}$ = Pathloss of UMa LOS outdoor scenario.

Note 4: $PL_{\mathrm{UMi-LOS}}$ = Pathloss of UMi-Street Canyon LOS outdoor scenario.

Note 5: Break point distance $d_{BP} = 2\pi\, h_{BS}\, h_{UT}\, f_c/c$, where f_c is the centre frequency in Hz, $c = 3.0 \times 10^8$ m/s is the propagation velocity in free space, and h_{BS} and h_{UT} are the antenna heights at the BS and the UT, respectively.

Note 6: f_c denotes the center frequency normalized by 1GHz, all distance related values are normalized by 1m, unless it is stated otherwise.

Pathloss equations for the different environments in 3GPP (continued). From [3GPP 38.901]. © 3GPP.

where p_i is the proportion of the i-th material, such that $\sum p_i = 1$, PL_{npi} is an additional loss due to non-perpendicular incidence on the building (not further specified in 38.901, and the models for the different building materials are

$$L_{glass} = 2 + 0.2f \qquad \text{for standard multi-pane glass} \tag{7.69}$$
$$L_{IIR} = 23 + 0.3f \qquad \text{for IIR (energy-saving) glass} \tag{7.70}$$
$$L_{concrete} = 5 + 4f \qquad \text{for concrete} \tag{7.71}$$
$$L_{wood} = 4.85 + 0.12f \qquad \text{for wood} \tag{7.72}$$
$$\tag{7.73}$$

where f is in GHz. For penetration through external walls, there is a low-loss and a high-loss model

$$PL_{tw,low-loss} = 5 - 10\log_{10}\left(0.3 \cdot 10^{\frac{-L_{\mathrm{glass}}}{10}} + 0.7 \cdot 10^{\frac{-L_{\mathrm{concrete}}}{10}}\right) \tag{7.74}$$
$$PL_{tw,high-loss} = 5 - 10\log_{10}\left(0.7 \cdot 10^{\frac{-L_{\mathrm{IIRglass}}}{10}} + 0.3 \cdot 10^{\frac{-L_{\mathrm{concrete}}}{10}}\right) \quad . \tag{7.75}$$

The standard deviation σ_P of the penetration loss is 4.4 dB in the low-loss model and 6.5 dB in the high-loss model. In either case, the indoor loss is modeled as half the pathloss of the value in dB obtained from the normal indoor model.

Penetration loss into a car is modeled as lognormal, with mean of 9 dB and standard deviation of 5 dB.

Preliminaries

The general approach of 3GPP to modeling of the small-scale statistics was described in Sec. 7.5.2, with a flow diagram Fig. 7.10. In a first step, the environment and the locations of N_{BS} BSs are chosen, and locations of the UEs are generated at random. One also needs to specify other user-specific quantities, such as their velocity vector $\mathbf{v}$, with its direction drawn from a uniform $[0, 360°)$ distribution, as well as the specifics of the UE antenna or antenna array such as array orientation, Ω_{UE}, etc. Figure 7.25 illustrates the various angle definitions. When multiple UEs are dropped, the geometry-related parameters are computed from the locations of these UEs. The statistical parameters described below are computed independently.

As a further preliminary note, the delay-spread σ_{DS}, BS angle-spread σ_{AS} and shadow fading σ_{SF} parameters[10] of the signal from a particular BS to a given user can be written as:

$$10\log_{10}(\sigma_{\mathrm{DS}}) = \mu_{\mathrm{DS}} + \varepsilon_{\mathrm{DS}} X_1 \tag{7.76}$$
$$10\log_{10}(\sigma_{\mathrm{AS}}) = \mu_{\mathrm{AS}} + \varepsilon_{\mathrm{AS}} X_2 \tag{7.77}$$
$$10\log_{10}(\sigma_{\mathrm{SF}}) = \varepsilon_{\mathrm{SF}} X_3 \tag{7.78}$$

In the above equations X_1, X_2, X_3 are zero-mean, unit-variance Gaussian random variables. μ_{DS} and μ_{AS} represent the median of the delay-spread and angle-spreads in dB. Similarly, the ε-coefficients are the log-normal standard deviations of each parameter. Note that for conciseness, we described the above equation only for three parameters; in the actual model there are 7 such correlated parameters (Rice factor, shadowing, delay spread, azimuth and elevation spreads at TX, and azimuth and elevation spreads at the RX. The correlation between those parameters is given by a correlation matrix $\mathbf{R}$, and correlated realizations of Gaussian variables can be obtained by pre-multiplying a vector of independent Gaussian variables with the square root of $\mathbf{R}$, where the square root can be obtained, e.g., from a Cholesky decomposition or an eigenvalue decomposition.

[10] We use here the notation σ_{DS} for the delay spread, instead of S_τ as in the remainder of the book, to retain consistency with the 3GPP report; similar for the angular spread and shadowing.

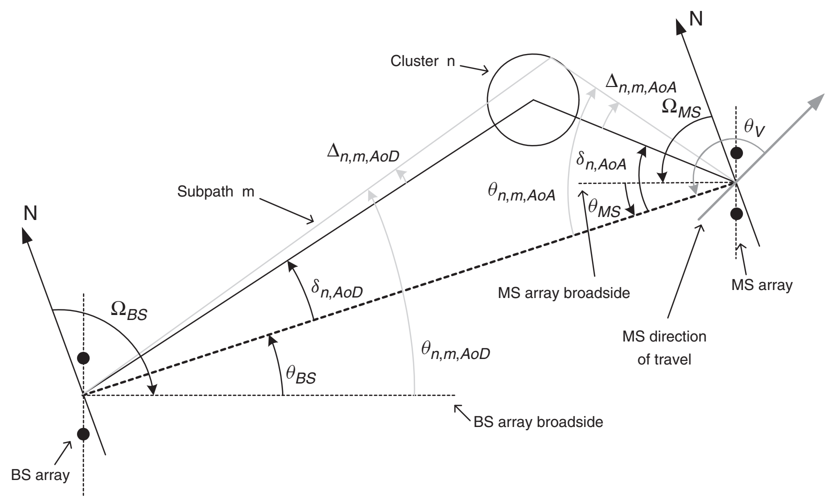

Figure 7.25: Angular variables definitions. Note that "MS" in this figure denotes the UE. From [Calcev et al. 2007] © IEEE.

We now describe the principle of generating fast fading coefficients for wideband time-varying MIMO channels with N_t transmit antennas and N_r receive antennas. The fast-fading coefficients for each of the N paths are constructed by the superposition of M individual sub-paths. The m-th component ($m = 1, \ldots, M$) is characterized by a relative angular offset to the mean AoD of the path at the BS, a relative angular offset to the mean AoA at the UE, a power and an overall phase. M is fixed to $M = 20$, and all sub-paths have the same power P_n/M. The sub-path delays are identical and equal to their corresponding path's delay. The overall phase of each subpath $\Phi_{n,m}$ is i.i.d., with a uniform distribution between 0 and 2π. The relative offset of the m-th subpath $\Delta_{n,m,AoD}$ at the BS, and $\Delta_{n,m,AoA}$ at the UE are deterministic and prescribed in a table. These offsets are chosen to result to the desired fixed per-path angle spreads. These per-path angle spreads are different from the narrowband angle spread σ_{AS} of the composite signal with N paths. Lastly, the path loss, based on the BS to UE distance and the log normal shadow fading, is applied to each of the sub-path powers of the channel model. The channel transfer function between any pair of TX and RX antenna elements can then be synthesized according to the principles of Sec. 6.7.4.

Simulation procedure

In this subsection we give the detailed simulation procedure to implement the above principles.

1. Select the number and locations of the BS(s) and the UE(s), their antenna patterns and configurations (including array features), and orientations in a global coordinate system. For the UE, furthermore select velocity vectors, according to probability density functions. For system evaluations within 3GPP, recommended values for these BS and UE parameters and distributions are prescribed in [3GPP 38.901]. Determine the directions of the direct geometrical line between BS and UE, which is the reference.

   Specify the environment, the carrier frequency, and the bandwidth.

2. Assign whether a UE has LOS or NLOS, according to the probabilities defined in Sec. 7.1.7; these states are uncorrelated between all UEs. Also specify whether the UEs are indoor or outdoor (or in vehicles); these are uncorrelated for different UEs, but the links from one UE to different BSs are all in the same state (e.g., UE is indoor for all links).

3. Calculate the pathloss according to the section *Pathloss* in this appendix.

4. Generate the large-scale parameters: delay spread, azimuth and elevation spreads at BS and UE, Rice factor, and shadow fading. All of those variables are modeled as lognormal, so that their logarithms are Gaussian variables, as discussed in the preliminaries. The realizations of azimuth spreads are limited to 104°, and for elevation to 52°. Furthermore, the parameters of different UEs connected to one BS (or equivalently, parameters for a single UE moving to different locations) are correlated. The correlation is modeled as exponentially decaying, with the $1/e$ decorrelation distance prescribed in the parameter tables provided below. Correlations of links from one UE to different BSs are neglected (though links from a UE to different sector-BSs at the same location are correlated).

5. Generate delays of the taps, either according to a uniform distribution, or - when the delay distribution is exponential - according to

$$\tau'_n = -r_{\mathrm{DS}}\sigma_{\mathrm{DS}} \ln z_n \quad n = 1, \ldots, N \tag{7.79}$$

where z_n $(n = 1, \ldots, N)$ are i.i.d. random variables with uniform distribution $U(0, 1)$ The τ'_n variables are then ordered so that $\tau'_{(N)} > \tau'_{(N-1)} > \cdots > \tau'_{(1)}$. Then their minimum is subtracted from all, i.e. $\tau_n = \left(\tau'_{(n)} - \tau'_{(1)}\right)$, with $n = 1, \ldots, N$, so that $\tau_N > \cdots > \tau_1 = 0$. In the case of LOS, each of the τ_n is to be divided by a factor

$$D = 0.7705 - 0.0433K + 0.0002K^2 + 0.000017K^3 \tag{7.80}$$

where K is the Rice factor in dB, as given in Table 7.12. Note, however, that for the generation of the tap powers (see next step) the *un*scaled τ_n are to be used.

6. Generate the average powers of the N taps as

$$P'_n = \exp\left[\frac{(1 - r_{\mathrm{DS}})\tau_n}{r_{\mathrm{DS}} \cdot \sigma_{\mathrm{DS}}}\right] \cdot 10^{-0.1u_n} \qquad n = 1, \ldots, N. \tag{7.81}$$

u_n for $n = 1, \ldots, N$ are the cluster shadowing terms that are lognormally distributed with shadowing standard deviation (in dB) equal to ξ. Average powers are then normalized, so that the total average power for all N paths is equal to unity. Each ray in a tap is assigned $1/M = 1/20$ of the tap power. In the case of LOS, an extra component with power $K_r/(K_r+1)$ is added, while the other components are further scaled by $1/(K_r + 1)$.

7. Generate the angles of arrival. The azimuth spectra are modeled as wrapped Gaussians, while the elevation spectra are modeled as Laplacian. In either case, the angles (η here denotes a generic angle) are computed in the Gaussian case as

$$\eta'_n = \frac{2(\sigma_{\mathrm{AS}}/1.4)\sqrt{-\ln(P_n/\max(P_n))}}{C} \tag{7.82}$$

and for the Laplacian case

$$\eta'_n = -\frac{\sigma_{\mathrm{AS}} \ln(P_n/\max(P_n))}{C} \tag{7.83}$$

The scaling constant C is a scaling factor related to the number of clusters, and is given in Tables 7.8 and7.9. In the case of LOS, the scaling constant has to be replaced by

$$C^{\mathrm{LOS}} = C\left(1.1035 - 0.028K - 0.002K^2 + 0.0001K^3\right) \tag{7.84}$$

$$C^{\mathrm{LOS}} = C\left(1.3086 + 0.0339K - 0.0077K^2 + 0.0002K^3\right) \tag{7.85}$$

for Gaussian and Laplacian spectrum, respectively.

The mean angles η_n are then computed as

$$\eta_n = X_n \eta_n' + Y_n + \eta_{\mathrm{LOS}} \tag{7.86}$$

where X_n is a random variable that equiprobably takes on the values $+1$ and -1, Y_n is a normally distributed random variable with standard deviation $\sigma_{\mathrm{AS}}/7$, and η_{LOS} is the arrival angle of the LOS component (which is known from gometrical considerations).

For the LOS case, η_n is normalized by subtracting $X_1\eta_1 + Y_1$ from every η_n; this enforces that the first cluster is seen in the direction η_{LOS}. The angles of the rays are computed by adding the offset angles (where the angles given in Table 7.10 have to be multiplied by the cluster angular spread) to the mean angles η_n of each cluster.

This procedure is applied to all angles, i.e., azimuth and elevation, at both BS and UE. Except that for the elevation at the BS, the offset is the ray offset multiplied by $(3/8)(10^{\mu_{\mathrm{ZSD}}})$.

8. Perform a random coupling of the ray AOAs and AODs within each cluster (or subcluster, see below). In other words, each cluster has only 20 rays, so that an AoD is uniquely coupled to an AoA, and this association is done at random. Azimuth and elevation angles are also randomly coupled.

9. In case polarization is required, for each ray, determine the cross-polarization ratio as $\kappa_{n,m} = 10^{X_{n,m}/10}$ where each $X_{n,m}$ is independently drawn from $\mathcal{N}(\mu_{\mathrm{XPR}}, \sigma_{\mathrm{XPR}})$.

10. For each ray, and each polarization (vv, vh, hv, hh) draw an initial random phase Φ from a uniform distribution in $[-\pi, \pi]$. In the LOS case, the phases are deterministic (see below).

11. For the two strongest taps, split the taps into "subclusters", according to Table 7.11, so that some of the taps have a delay offset compared to the nominal computed delays. For the weaker clusters, no such splitting is done.

12. Compute the polarized impulse responses at the antenna elements as

$$\begin{aligned} h_{u,s,n}^{\mathrm{NLOS}}(t) \quad = \quad & \sqrt{\frac{P_n}{M}} \sum_{m=1}^{M} \left(e^{j<\mathbf{k}_{0,m,n}^{\mathrm{RX}}\mathbf{v}>} \right) \cdot \\ & \tilde{\mathbf{G}}_{RX}(\phi_{n,m}^{\mathrm{RX}}, \theta_{n,m}^{\mathrm{RX}}) \cdot \begin{matrix} \exp\left(j\Phi_{n,m}^{\mathrm{vv}}\right) & \kappa^{-1/2}\exp\left(j\Phi_{n,m}^{\mathrm{vh}}\right) \\ \kappa^{-1/2}\exp\left(j\Phi_{n,m}^{\mathrm{hv}}\right) & \exp\left(j\Phi_{n,m}^{\mathrm{hh}}\right) \end{matrix} \\ & \tilde{\mathbf{G}}_{TX}(\phi_{n,m}^{\mathrm{TX}}, \theta_{n,m}^{\mathrm{TX}}) \cdot e^{j<\mathbf{k}_{0,m,n}^{\mathrm{RX}}\mathbf{r}_u>} e^{j<\mathbf{k}_{0,m,n}^{\mathrm{TX}}\mathbf{r}_s>} \end{aligned} \tag{7.87}$$

where $k_{0,m,n}^{\mathrm{RX}}$ is the wave vector in the direction of the DOA of the mth ray of the nth tap, u indexes the RX antenna elements, s the TX antenna elements, $\mathbf{r}_u$ is the location vector of the uth RX antenna element, and similarly for the TX antenna elements. Due to the different definition of the DoA, the sign of the $e^{j<\mathbf{k}_{0,m,n}^{\mathrm{RX}}\mathbf{r}_u>}$ is different from (6.70). *phi* and *theta* are azimuth and elevation. Note that the $\tilde{\mathbf{G}}$ are the (vector) complex array pattern that include the phase shifts due to the displacement of the antenna elements, see Chapter 8. The equations in [3GPP 38.901] assume an *element* pattern F, so that the equation for the total signal includes an explicit phase shift term for both Tx and Rx element displacement.

In the case of LOS, add a component, so that the overall impulse response in the LOS situation is

$$\begin{aligned} h_{u,s,n}^{\mathrm{LOS}}(t) \quad = \quad & \sqrt{\frac{1}{1+K}} h_{u,s,n}^{\mathrm{NLOS}}(t) + \delta(n-1)\sqrt{\frac{K}{1+K}} \left(e^{j<\mathbf{k}_{0,LOS}^{\mathrm{RX}}\mathbf{v}>} \right) \cdot \\ & \tilde{\mathbf{G}}_{RX}(\phi_{LOS}^{\mathrm{RX}}, \theta_{LOS}^{\mathrm{RX}}) \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tilde{\mathbf{G}}_{TX}(\phi_{LOS}^{\mathrm{TX}}, \theta_{LOS}^{\mathrm{TX}}) \end{aligned} \tag{7.88}$$

No. clusters	4	5	8	10	11	12	14	15	16	19	20	25
C	0.779	0.860	1.018	1.090	1.123	1.146	1.190	1.211	1.226	1.273	1.289	1.358

Table 7.8: Scaling factors for generation of the azimuth angles

Number of Clusters	8	10	11	12	15	19	20	25
C	0.889	0.957	1.031	1.104	1.1088	1.184	1.178	1.282

Table 7.9: Scaling factors for generation of the elevation angles

13. Apply the pathloss and (bulk) shadowing to the overall impulse response.

enforcement of spatial consistency.

Ray number, m	**Basis vector of offset angles, α_{ms}**
1,2	± 0.0447
3,4	± 0.1413
5,6	± 0.2492
7,8	± 0.3715
9,10	± 0.5129
11,12	± 0.6797
13,14	± 0.8844
15,16	± 1.1481
17,18	± 1.5195
19,20	± 2.1551

Table 7.10: Ray offset angles within a cluster, given for 1 degree rms angle spread

Sub-cluster No.	**Mapping to rays**	**Power**	**Delay offset**
1	1, 2, 3, 4, 5, 6, 7, 8, 19, 20	10/20	0
2	9, 10, 11, 12, 17, 18	6/20	$1.28c_{\mathrm{DS}}$
3	13, 14, 15, 16	4/20	$2.56c_{\mathrm{DS}}$

Table 7.11: Sub-cluster information for intra cluster delay spread cluster

Scenarios		UMi - Street Canyon			UMa		
		LOS	NLOS	O2I	LOS	NLOS	O2I
Delay spread (DS) lgDS=log$_{10}$(DS/1s)	μ_{lgDS}	$-0.24 \log_{10}(1+f_c) - 7.14$	$-0.24 \log_{10}(1+f_c) - 6.83$	-6.62	$-6.955 - 0.0963 \log_{10}(f_c)$	$-6.28 - 0.204 \log_{10}(f_c)$	-6.62
	σ_{lgDS}	0.38	$0.16 \log_{10}(1+f_c) + 0.28$	0.32	0.66	0.39	0.32
AOD spread (ASD) lgASD=log$_{10}$(ASD/1°)	μ_{lgASD}	$-0.05 \log_{10}(1+f_c) + 1.21$	$-0.23 \log_{10}(1+f_c) + 1.53$	1.25	$1.06 + 0.1114 \log_{10}(f_c)$	$1.5 - 0.1144 \log_{10}(f_c)$	1.25
	σ_{lgASD}	0.41	$0.11 \log_{10}(1+f_c) + 0.33$	0.42	0.28	0.28	0.42
AOA spread (ASA) lgASA=log$_{10}$(ASA/1°)	μ_{lgASA}	$-0.08 \log_{10}(1+f_c) + 1.73$	$-0.08 \log_{10}(1+f_c) + 1.81$	1.76	1.81	$2.08 - 0.27 \log_{10}(f_c)$	1.76
	σ_{lgASA}	$0.014 \log_{10}(1+f_c) + 0.28$	$0.05 \log_{10}(1+f_c) + 0.3$	0.16	0.20	0.11	0.16
ZOA spread (ZSA) lgZSA=log$_{10}$(ZSA/1°)	μ_{lgZSA}	$-0.1 \log_{10}(1+f_c) + 0.73$	$-0.04 \log_{10}(1+f_c) + 0.92$	1.01	0.95	$-0.3236 \log_{10}(f_c) + 1.512$	1.01
	σ_{lgZSA}	$-0.04 \log_{10}(1+f_c) + 0.34$	$-0.07 \log_{10}(1+f_c) + 0.41$	0.43	0.16	0.16	0.43
Shadow fading (*SF*) [dB]	σ_{SF}	See Table 7.4.1-1	See Table 7.4.1-1	7	See Table 7.4.1-1	See Table 7.4.1-1	7
K-factor (*K*) [dB]	μ_K	9	N/A	N/A	9	N/A	N/A
	σ_K	5	N/A	N/A	3.5	N/A	N/A
Cross-Correlations	*ASD* vs *DS*	0.5	0	0.4	0.4	0.4	0.4
	ASA vs *DS*	0.8	0.4	0.4	0.8	0.6	0.4
	ASA vs *SF*	-0.4	-0.4	0	-0.5	0	0
	ASD vs *SF*	-0.5	0	0.2	-0.5	-0.6	0.2
	DS vs *SF*	-0.4	-0.7	-0.5	-0.4	-0.4	-0.5
	ASD vs *ASA*	0.4	0	0	0	0.4	0
	ASD vs K	-0.2	N/A	N/A	0	N/A	N/A
	ASA vs K	-0.3	N/A	N/A	-0.2	N/A	N/A
	DS vs K	-0.7	N/A	N/A	-0.4	N/A	N/A
	SF vs K	0.5	N/A	N/A	0	N/A	N/A
Cross-Correlations [1)]	*ZSD* vs *SF*	0	0	0	0	0	0
	ZSA vs *SF*	0	0	0	-0.8	-0.4	0
	ZSD vs *K*	0	N/A	N/A	0	N/A	N/A
	ZSA vs *K*	0	N/A	N/A	0	N/A	N/A
	ZSD vs *DS*	0	-0.5	-0.6	-0.2	-0.5	-0.6
	ZSA vs *DS*	0.2	0	-0.2	0	0	-0.2

Large-scale parameters for the different environments in 3GPP. From [3GPP 38.901]. © 3GPP.

	ZSD vs *ASD*	0.5	0.5	-0.2	0.5	0.5	-0.2
	ZSA vs *ASD*	0.3	0.5	0	0	-0.1	0
	ZSD vs *ASA*	0	0	0	-0.3	0	0
	ZSA vs *ASA*	0	0.2	0.5	0.4	0	0.5
	ZSD vs *ZSA*	0	0	0.5	0	0	0.5
Delay scaling parameter r_τ		3	2.1	2.2	2.5	2.3	2.2
XPR [dB]	μ_{XPR}	9	8.0	9	8	7	9
	σ_{XPR}	3	3	5	4	3	5
Number of clusters N		12	19	12	12	20	12
Number of rays per cluster M		20	20	20	20	20	20
Cluster *DS* (c_{DS}) in [ns]		5	11	11	max(0.25, 6.5622 -3.4084 $\log_{10}(f_c)$)	max(0.25, 6.5622 -3.4084 $\log_{10}(f_c)$)	11
Cluster *ASD* (c_{ASD}) in [deg]		3	10	5	5	2	5
Cluster *ASA* (c_{ASA}) in [deg]		17	22	8	11	15	8
Cluster *ZSA* (c_{ZSA}) in [deg]		7	7	3	7	7	3
Per cluster shadowing std ζ [dB]		3	3	4	3	3	4
Correlation distance in the horizontal plane [m]	*DS*	7	10	10	30	40	10
	ASD	8	10	11	18	50	11
	ASA	8	9	17	15	50	17
	SF	10	13	7	37	50	7
	K	15	N/A	N/A	12	N/A	N/A
	ZSA	12	10	25	15	50	25
	ZSD	12	10	25	15	50	25

f_c is carrier frequency in GHz; d_{2D} is BS-UT distance in km.

NOTE 1: *DS* = rms delay spread, *ASD* = rms azimuth spread of departure angles, *ASA* = rms azimuth spread of arrival angles, *ZSD* = rms zenith spread of departure angles, *ZSA* = rms zenith spread of arrival angles, *SF* = shadow fading, and *K* = Ricean K-factor.

NOTE 2: The sign of the shadow fading is defined so that positive SF means more received power at UT than predicted by the path loss model.

NOTE 3: All large scale parameters are assumed to have no correlation between different floors.

NOTE 4: The following notation for mean (μ_{lgX}=mean{$\log_{10}$(X) }) and standard deviation (σ_{lgX}=std{$\log_{10}$(X) }) is used for logarithmized parameters X.

NOTE 5: For all considered scenarios the AOD/AOA distributions are modelled by a wrapped Gaussian distribution, the ZOD/ZOA distributions are modelled by a Laplacian distribution and the delay distribution is modelled by an exponential distribution.

NOTE 6: For UMa and frequencies below 6 GHz, use f_c = 6 when determining the values of the frequency-dependent LSP values

NOTE 7: For UMi and frequencies below 2 GHz, use f_c = 2 when determining the values of the frequency-dependent LSP values

Large-scale parameters for the different environments in 3GPP (continued). From [3GPP 38.901]. © 3GPP.

Scenarios		RMa			Indoor-Office	
		LOS	NLOS	O2I	LOS	NLOS
Delay spread (DS) $\lg DS=\log_{10}(DS/1s)$	$\mu_{\lg DS}$	-7.49	-7.43	-7.47	$-0.01 \log_{10}(1+f_c) - 7.692$	$-0.28 \log_{10}(1+f_c) - 7.173$
	$\sigma_{\lg DS}$	0.55	0.48	0.24	0.18	$0.10 \log_{10}(1+f_c) + 0.055$
AOD spread (ASD) $\lg ASD=\log_{10}(ASD/1°)$	$\mu_{\lg ASD}$	0.90	0.95	0.67	1.60	1.62
	$\sigma_{\lg ASD}$	0.38	0.45	0.18	0.18	0.25
AOA spread (ASA) $\lg ASA=\log_{10}(ASA/1°)$	$\mu_{\lg ASA}$	1.52	1.52	1.66	$-0.19 \log_{10}(1+f_c) + 1.781$	$-0.11 \log_{10}(1+f_c) + 1.863$
	$\sigma_{\lg ASA}$	0.24	0.13	0.21	$0.12 \log_{10}(1+f_c) + 0.119$	$0.12 \log_{10}(1+f_c) + 0.059$
ZOA spread (ZSA) $\lg ZSA=\log_{10}(ZSA/1°)$	$\mu_{\lg ZSA}$	0.47	0.58	0.93	$-0.26 \log_{10}(1+f_c) + 1.44$	$-0.15 \log_{10}(1+f_c) + 1.387$
	$\sigma_{\lg ZSA}$	0.40	0.37	0.22	$-0.04 \log_{10}(1+f_c) + 0.264$	$-0.09 \log_{10}(1+f_c) + 0.746$
Shadow fading (SF) [dB]	σ_{SF}	See Table 7.4.1-1		8	See Table 7.4.1-1	
K-factor (K) [dB]	μ_K	7	N/A	N/A	7	N/A
	σ_K	4	N/A	N/A	4	N/A
Cross-Correlations	*ASD* vs *DS*	0	-0.4	0	0.6	0.4
	ASA vs *DS*	0	0	0	0.8	0
	ASA vs *SF*	0	0	0	–0.5	–0.4
	ASD vs *SF*	0	0.6	0	–0.4	0
	DS vs *SF*	-0.5	-0.5	0	–0.8	–0.5
	ASD vs *ASA*	0	0	-0.7	0.4	0
	ASD vs K	0	N/A	N/A	0	N/A
	ASA vs K	0	N/A	N/A	0	N/A
	DS vs K	0	N/A	N/A	-0.5	N/A
	SF vs K	0	N/A	N/A	0.5	N/A
Cross-Correlations [1]	*ZSD* vs *SF*	0.01	-0.04	0	0.2	0
	ZSA vs *SF*	-0.17	-0.25	0	0.3	0
	ZSD vs *K*	0	N/A	N/A	0	N/A
	ZSA vs *K*	-0.02	N/A	N/A	0.1	N/A
	ZSD vs *DS*	-0.05	-0.10	0	0.1	-0.27
	ZSA vs *DS*	0.27	-0.40	0	0.2	-0.06
	ZSD vs *ASD*	0.73	0.42	0.66	0.5	0.35
	ZSA vs *ASD*	-0.14	-0.27	0.47	0	0.23
	ZSD vs *ASA*	-0.20	-0.18	-0.55	0	-0.08
	ZSA vs *ASA*	0.24	0.26	-0.22	0.5	0.43
	ZSD vs *ZSA*	-0.07	-0.27	0	0	0.42
Delay scaling parameter r_τ		3.8	1.7	1.7	3.6	3
XPR [dB]	μ_{XPR}	12	7	7	11	10
	σ_{XPR}	4	3	3	4	4
Number of clusters N		11	10	10	15	19
Number of rays per cluster M		20	20	20	20	20
Cluster *DS* (c_{DS}) in [ns]		N/A	N/A	N/A	N/A	N/A
Cluster *ASD* (c_{ASD}) in [deg]		2	2	2	5	5
Cluster *ASA* (c_{ASA}) in [deg]		3	3	3	8	11
Cluster *ZSA* (c_{ZSA}) in [deg]		3	3	3	9	9
Per cluster shadowing std ζ [dB]		3	3	3	6	3
Correlation distance in the horizontal plane [m]	*DS*	50	36	36	8	5
	ASD	25	30	30	7	3
	ASA	35	40	40	5	3
	SF	37	120	120	10	6
	K	40	N/A	N/A	4	N/A
	ZSA	15	50	50	4	4
	ZSD	15	50	50	4	4
f_c is carrier frequency in GHz; d_{2D} is BS-UT distance in km. For InH and frequencies below 6 GHz, use $f_c = 6$ when determining the values of the frequency-dependent LSP values						

Large-scale parameters for the different environments in 3GPP (continued). From [3GPP 38.901]. © 3GPP.

Scenarios		InF	
		LOS	NLOS
Delay spread (DS) lgDS=$\log_{10}$(DS/1s) 4)	μ_{lgDS}	$\log_{10}$(26(V/S)+14)-9.35	$\log_{10}$(30(V/S)+32)-9.44
	σ_{lgDS}	0.15	0.19
AOD spread (ASD) lgASD=$\log_{10}$(ASD/1°)	μ_{lgASD}	1.56	1.57
	σ_{lgASD}	0.25	0.2
AOA spread (ASA) lgASA=$\log_{10}$(ASA/1°)	μ_{lgASA}	-0.18*log10(1+fc) + 1.78	1.72
	σ_{lgASA}	0.12*log10(1+fc) + 0.2	0.3
ZOA spread (ZSA) lgZSA=$\log_{10}$(ZSA/1°)	μ_{lgZSA}	-0.2*log10(1+fc) + 1.5	-0.13*log10(1+fc) + 1.45
	σ_{lgZSA}	0.35	0.45
Shadow fading (SF) [dB]	σ_{SF}	Specified as part of path loss models	
K-factor (*K*) [dB]	μ_K	7	N/A
	σ_K	8	N/A
Cross-Correlations	*ASD* vs *DS*	0	0
	ASA vs *DS*	0	0
	ASA vs *SF*	0	0
	ASD vs *SF*	0	0
	DS vs *SF*	0	0
	ASD vs *ASA*	0	0
	ASD vs *K*	-0.5	N/A
	ASA vs *K*	0	N/A
	DS vs *K*	-0.7	N/A
	SF vs *K*	0	N/A
Cross-Correlations 1)	*ZSD* vs *SF*	0	0
	ZSA vs *SF*	0	0
	ZSD vs *K*	0	N/A
	ZSA vs *K*	0	N/A
	ZSD vs *DS*	0	0
	ZSA vs *DS*	0	0
	ZSD vs *ASD*	0	0
	ZSA vs *ASD*	0	0
	ZSD vs *ASA*	0	0
	ZSA vs *ASA*	0	0
	ZSD vs *ZSA*	0	0
Delay scaling parameter r_τ		2.7	3
XPR [dB]	μ_{XPR}	12	11
	σ_{XPR}	6	6
Number of clusters N		25	25
Number of rays per cluster M		20	20
Cluster *DS* (c_{DS}) in [ns]		N/A	N/A
Cluster *ASD* (c_{ASD}) in [deg]		5	5
Cluster *ASA* (c_{ASA}) in [deg]		8	8
Cluster *ZSA* (c_{ZSA}) in [deg]		9	9
Per cluster shadowing std ζ [dB]		4	3
Correlation distance in the horizontal plane [m] 5)	*DS*	10	10
	ASD	10	10
	ASA	10	10
	SF	10	10
	K	10	N/A
	ZSA	10	10
	ZSD	10	10
f_c is carrier frequency in GHz; d_{2D} is BS-UT distance in km. NOTE 4: V = hall volume in m^3, S = total surface area of hall in m^2 (walls+floor+ceiling) NOTE 5: When the simulation scenario is small it is recommended to consider methods to improve the statistical confidence, such as using multiple random seeds in simulations.			

Large-scale parameters for the different environments in 3GPP (continued). From [3GPP 38.901].

Scenarios - UMa		**LOS**	**NLOS**
ZOD spread (ZSD) lgZSD=$\log_{10}$(ZSD/1°)	μ_{lgZSD}	max[-0.5, -2.1(d_{2D}/1000) -0.01 (h_{UT} - 1.5)+0.75]	max[-0.5, -2.1(d_{2D}/1000) -0.01(h_{UT} - 1.5)+0.9]
	σ_{lgZSD}	0.40	0.49
ZOD offset	$\mu_{offset,ZOD}$	0	$e(f_c)$-10^{$a(f_c)$ $\log_{10}$(max($b(f_c)$, d_{2D})) +$c(f_c)$ -0.07(h_{UT}-1.5)}
Note: For NLOS ZOD offset: $a(f_c)$ = 0.208$\log_{10}(f_c)$- 0.782; $b(f_c)$ = 25; $c(f_c)$ = -0.13$\log_{10}(f_c)$+2.03; $e(f_c)$ = 7.66$\log_{10}(f_c)$-5.96.			

Scenarios - UMi		**LOS**	**NLOS**
ZOD spread (ZSD) lgZSD=$\log_{10}$(ZSD/1°)	μ_{lgZSD}	max[-0.21, -14.8(d_{2D}/1000) + 0.01|h_{UT}-h_{BS}| + 0.83]	max[-0.5, -3.1(d_{2D}/1000) + 0.01 max(h_{UT}-h_{BS},0) +0.2]
	σ_{lgZSD}	0.35	0.35
ZOD offset	$\mu_{offset,ZOD}$	0	-10^{-1.5$\log_{10}$(max(10, d_{2D}))+3.3}

Scenarios - RMa		**LOS**	**NLOS**
ZOD spread (ZSD) lgZSD=$\log_{10}$(ZSD/1°)	μ_{lgZSD}	max[-0.21, -14.8(d_{2D}/1000) + 0.01|h_{UT}-h_{BS}| + 0.83]	max[-0.5, -3.1(d_{2D}/1000) + 0.01 max(h_{UT}-h_{BS},0) +0.2]
	σ_{lgZSD}	0.35	0.35
ZOD offset	$\mu_{offset,ZOD}$	0	-10^{-1.5$\log_{10}$(max(10, d_{2D}))+3.3}

Scenarios		**LOS**	**NLOS**
ZOD spread (ZSD) lgZSD=$\log_{10}$(ZSD/1°)	μ_{lgZSD}	-1.43 $\log_{10}$(1+ f_c)+2.228	1.08
	σ_{lgZSD}	0.13 $\log_{10}$(1+f_c)+0.30	0.36
ZOD offset	$\mu_{offset,ZOD}$	0	0

Scenarios		**LOS**	**NLOS**
ZOD spread (ZSD) lgZSD=$\log_{10}$(ZSD/1°)	μ_{lgZSD}	1.35	1.2
	σ_{lgZSD}	0.35	0.55
ZOD offset	$\mu_{offset,ZOD}$	0	0

Parameters for ray offsets for the different environments in 3GPP. From [3GPP 38.901]. © 3GPP.

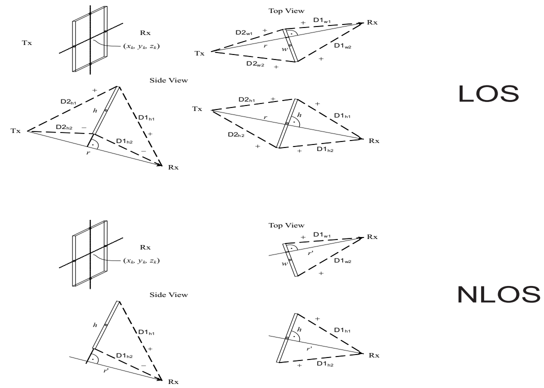

Figure 7.26: 2". From [3GPP 38901]

Shaodwing by moving objects

The 3GPP model also contains a mathematical description of how rays are shadowed off when obstructed by objects such as humans, cars, etc. There are two models, A and B. In model B (the only one we describe here), the objects are approximated as screens, and approximate diffraction equations are used to compute the attenuation.

The geometry of the model and its parameters $D1_{ij}$, $D1_{ij}$are shown in Fig. 7.26. The shadowing loss is determined by diffracted fields from four edges of the screen as

$$L_{\mathrm{dB}} = -20\log_{10}\left(1 - \left(F_{h_1} + F_{h_2}\right)\left(F_{w_1} + F_{w_2}\right)\right) \tag{7.89}$$

where

$$F_{h_1|h_2|w_1|w_2} = \begin{cases} \dfrac{\tan^{-1}\left(\pm\frac{\pi}{2}\sqrt{\frac{\pi}{\lambda}\left(\mathrm{D1}_{h_1|h_2|w_1|w_2}+\mathrm{D2}_{h_1|h_2|w_1|w_2}-r\right)}\right)}{\pi} & ,\mathrm{fordirectpathinLOS} \\ \dfrac{\tan^{-1}\left(\pm\frac{\pi}{2}\sqrt{\frac{\pi}{\lambda}\left(\mathrm{D1}_{h_1|h_2|w_1|w_2}-r'\right)}\right)}{\pi} & ,\mathrm{forallotherpaths} \end{cases} \tag{7.90}$$

The screen is always rotated to be perpendicular to the sub-path. Losses from multiple screens are added up on a dB scale (for a discussion of the validity of this assumption, see Sec. 4.3).

Further generalizations, such as inclusion of oxygen absorption, simulations for large arrays and/or large bandwidth, spatial consistency, dual mobility, and multi-band simulations are also provided in [3GPP 38.901].

Applicability in various environments

The 3GPP channel model was designed by a standardization body *for the purpose of comparing different system proposals*. As mentioned in Chapter 7, this implies that it is *not intended for absolute performance assessment*. It makes a large number of simplifications (to reduce computation time) and assumptions (not necessarily in agreement with measurement results) that will result in deviations of absolute performance prediction from measured realities. This is compounded by the long evolution of the model that resulted in the requirement of backward compatibility with previous versions. This is particularly true for modeling at higher frequencies, where having a simple generalization of the below-6 GHz was considered paramount by 3GPP. Adequate performance of a communication system over a simulated 3GPP channel is thus only a *relative* measure, and cannot be used to conclude adequate *absolute* performance.

Appendix 7.E: The 802.15.4a UWB Channel Model

The IEEE 802.15.4a group defined ultrawideband channel models: one model for office environments below 1 GHz, one model for body-area networks, and one group of models for indoor residential, indoor office, outdoor, and industrial environments. In this appendix, we only describe the last group of models, since it is the most frequently used. The other models are discussed in [Molisch et al. 2004].

We assume that the path gain as a function of the distance and frequency can be written as a product of the terms

$$G(f,d) = G(f) \cdot G(d). \tag{7.91}$$

where the frequency dependence of the channel path gain is

$$\sqrt{G(f)} \propto f^{-\kappa} \tag{7.92}$$

where κ is the frequency dependency decaying factor. The distance dependence is given by the standard power-decay law Eq. (7.8) with a breakpoint distance of 1 m, combined with lognormal shadowing (Sec. 5.8).

The power delay profile is given by a modified Saleh-Valenzuela model (Sec. 7.2.3)

$$h_{\mathrm{discr}}(t) = \sum_{m=0}^{M} \sum_{k=0}^{K} a_{k,m} \exp(j\phi_{k,m}) \delta(t - T_m - \tau_{k,m}), \tag{7.93}$$

where the phases $\phi_{k,m}$ are uniformly distributed in $[0,2\pi)$ with a number of clusters that is Poisson distributed.

$$pdf_M(M) = \frac{(\overline{M})^M \exp(-\overline{M})}{M!} \tag{7.94}$$

so that the mean $\overline{L}$ completely characterizes the distribution.

By definition, we have $\tau_{0,m} = 0$. The distributions of the cluster arrival times are given by a Poisson process

$$p(T_m|T_{m-1}) = \Lambda_m \exp\left[-\Lambda_m(T_m - T_{m-1})\right], \; m > 0 \tag{7.95}$$

where Λ_m is the cluster arrival rate (assumed to be independent of l). The classical SV model also uses a Poisson process for the ray arrival times. Due to the discrepancy in the fitting for the indoor residential, indoor office, and outdoor environments, we model ray arrival times with a mixture of two Poisson processes as

$$\begin{aligned} &p\left(\tau_{k,m}|\tau_{(k-1),m}\right) = \beta\lambda_1 \exp\left[-\lambda_1\left(\tau_{k,m} - \tau_{(k-1),m}\right)\right] \\ &+ (1-\beta)\,\lambda_2 \exp\left[-\lambda_2\left(\tau_{k,m} - \tau_{(k-1),m}\right)\right], \quad k > 0 \end{aligned} \tag{7.96}$$

where β is the mixture probability, while λ_1 and λ_2 are the ray arrival rates.

For some environments, most notably the industrial environment, a "dense" arrival of MPCs was observed, i.e., each resolvable delay bin contains significant energy. In that case, the concept of ray arrival rates loses its meaning, and a realization of the impulse response based on a tapped delay line model with regular tap spacings is to be used.

The next step is the determination of the cluster powers and cluster shapes. The PDP (mean power of the different paths) is exponential within each cluster

$$E\{|a_{k,m}|^2\} \propto \Omega_m \exp(-\tau_{k,m}/\gamma_m) \tag{7.97}$$

where Ω_m is the integrated energy of the mth cluster, and γ_m is the intra-cluster decay time constant.

Another important deviation from the classical SV model is that we find the cluster decay time to depend on the arrival time of the cluster. In other words, the larger the delay, the larger the decay time of the cluster. A linear dependence

$$\gamma_m \propto k_\gamma T_m + \gamma_0 \tag{7.98}$$

Table 7.16: Parameters for channel models CM 1 and CM 2 (residential)

Residential valid range of d	LOS $7-20$ m	NLOS $7-20$ m
Path gain		
G_0 [dB]	-43.9	-48.7
n	1.79	4.58
$S[dB]$	2.22	3.51
κ	1.12	1.53
Power delay profile		
$\overline{L}$	3	3.5
Λ [1/ns]	0.047	0.12
λ_1, λ_2 [1/ns], β	$1.54, 0.15, 0.095$	$1.77, 0.15, 0.045$
Γ [ns]	22.61	26.27
k_γ	0	0
γ_0 [ns]	12.53	17.50
σ_{cluster} [dB]	2.75	2.93
Small-scale fading		
m_0 [dB]	0.67	0.69
$\widehat{m}_0$ [dB]	0.28	0.32
$\widetilde{m}_0$	NA	NA

where k_γ describes the increase of the decay constant with delay, gives good agreement with measurement values.

The energy of the mth cluster, normalized to γ_m, and averaged over the cluster shadowing and the small-scale fading, follows in general an exponential decay

$$10\log_{10}(\Omega_m) = 10\log_{10}(\exp(-T_m/\Gamma)) + M_{\text{cluster}} \tag{7.99}$$

where M_{cluster} is a normally distributed variable with standard deviation σ_{cluster}.

For the non-LOS (NLOS) case of some environments (office and industrial), the above description is not a good model for the PDP. Rather, only a single cluster, whose power first increases (with increasing delay), goes through a maximum, and then decreases again, was observed experimentally. Such a shape of the PDP can significantly influence the ranging and geolocation performance of UWB systems, as ranging algorithms require the identification of the first (not the strongest) MPC.

The following functional fit is used in the 802.15.4a model:

$$E\{|a_{k,1}|^2\} \propto (1 - \chi \cdot \exp(-\tau_{k,l}/\gamma_{\text{rise}})) \cdot \exp(-\tau_{k,l}/\gamma_1). \tag{7.100}$$

Here, the parameter χ describes the attenuation of the first component, the parameter γ_{rise} determines how fast the PDP increases to its local maximum, and γ_1 determines the decay at later times.

For the small-scale fading, Nakagami fading is assumed, with the m-factor modeled as a lognormally fading random variable, with mean and standard deviation (in dB) as

$$\mu_{\text{m}} = \quad m_0 + k_{\text{m}}\tau \tag{7.101}$$

$$\sigma_{\text{m}} = \quad \hat{m}_0 + \hat{k}_{\text{m}}\tau \tag{7.102}$$

The numerical values of the parameters used in Eq. (7.91)–(7.100) are specified, for different environments, in Tables 7.16–7.19.

Table 7.17: Parameters for channel models CM 3 and CM 4 (office)

Office valid range of d	LOS $3 - 28$ m	NLOS $3 - 28$ m
Path gain		
n	1.63	3.07
σ_S	1.9	3.9
G_0 [dB]	-35.4	-59.9
κ	0.03	0.71
Power delay profile		
$\overline{L}$	5.4	1
Λ [1/ns]	0.016	NA
λ_1, λ_2 [1/ns], β	$0.19, 2.97, 0.0184$	NA
Γ [ns]	14.6	NA
k_γ	0	NA
γ_0 [ns]	6.4	NA
σ_{cluster} [dB]	3	NA
Small-scale fading		
m_0	0.42 dB	0.50 dB
$\widehat{m}_0$	0.31	0.25
$\widetilde{m}_0$	NA	NA
χ	NA	0.86
γ_{rise}	NA	15.21
γ_1	NA	11.84

Table 7.18: Parameters for channel models CM 5, CM 6, and CM 9 (outdoor LOS, outdoor NLOS, and farm environments)

Outdoor valid range of d	LOS $5 - 17$ m	NLOS $5 - 17$ m	Farm
Path gain			
n	1.76	2.5	1.58
σ_S	0.83	2	3.96
G_0	-45.6	-73.0	-48.96
κ	0.12	0.13	0
Power delay profile			
$\overline{L}$	13.6	10.5	3.31
Λ [1/ns]	0.0048	0.0243	0.0305
λ_1 [1/ns]	0.27	0.15	$0.0225, 0, 0$
λ_2 [1/ns]	2.41	1.13	
β	0.0078	0.062	
Γ [ns]	31.7	104.7	56
k_γ	0	0	0
γ_0 [ns]	3.7	9.3	0.92
σ_{cluster} [dB]	3	3	3
Small-scale fading			
m_0	0.77 dB	0.56 dB	4.1 dB
$\widehat{m}_0$	0.78	0.25	2.5 dB
$\widetilde{m}_0$	NA	NA	0

Table 7.19: Parameters for channel models CM 7 and CM 8 (industrial)

Industrial	LOS	NLOS
valid range of d	$2-8$ m	$2-8$ m
Path gain		
n	1.2	2.15
σ_{S} [dB]	6	6
G_0 [dB]	-56.7	-56.7
κ	-1.103	-1.427
Power delay profile		
$\overline{L}$	4.75	1
Λ [1/ns]	0.0709	NA
λ [1/ns]	NA	NA
Γ	13.47	NA
k_γ	0.926	NA
γ_0	0.651	NA
$\sigma_{\mathrm{cluster}}$ [dB]	4.32	NA
Small-scale fading		
m_0	0.36 dB	0.30 dB
$\widehat{m}_0$	1.13	1.15
$\widetilde{m}_0$ dB	12.99	
χ	NA	1
γ_{rise} [ns]	NA	17.35
γ_1 [ns]	NA	85.36

Appendix 7.F: The COST 259/273/2100 Channel Model

The COST 259/273/2100 is a geometry-based stochastic channel model (GSCM) that contains many advanced features. In contrast to the standardized models of 3GPP and IEEE, it does not have a unique implementation, as it is not designed for a specific bandwidth, but rather describes a general (continuous) setup that can be implemented by users according to their requirements.

We describe in the following first the original COST 259 model, and then mention the extensions provided by COST 273 and 2100, respectively. The description will focus on macrocells, which are the most completely parameterized.

The COST 259 model - General Aspects

Radio Environments (REs)

Macrocells are characterized by relative distances between BS and UE in the km-range, and a marked difference between the height of BS and UE. Thus, the immediate surroundings of BS and UE have different characteristics, leading to different angular characteristics of the channel. In micro- and picocells, on the other hand, BS and UE are mostly co-located in the same environment of scatterers.

Micro- and picocells are distinguished by the fact that picocell BSs are located indoors, while for microcells, they are located outdoors. In the scenarios described by the COST259DCM, the microcell UE is also outdoors, while the picocell UE is indoors. Thus, "penetration" scenarios (coverage of a building by an outdoor BS, or coverage of an office courtyard by an indoor BS) are not specified explicitly, though suggestions on how to derive them are given.

Macrocells distinguish between

GTU The generalized *Typical Urban* RE: Cities and towns where the buildings have nearly uniform height and density.

GBU The generalized *Bad Urban* RE: Cities with distinctly nonuniform building heights or densities, such as high-rise metropolitan centers and cities with large open areas such as rivers, lakes or parks.

GRA The generalized *Rural Area* RE: Environments with only few buildings, such as farmlands, fields and forests.

GHT The generalized *Hilly Terrain* RE: Similar to GRA, but with large height variations such as hills or mountains.

The REs are defined by their common propagation properties, but basically no statement is made how long a UE resides in one RE. In a macrocell, this duration is typically very long. In most cases a UE will remain in the same RE for the whole call duration. For micro- and picocells, transitions from one RE to another (e.g. from a street to an open place; from the office out into the corridor) can easily appear during a period of interest for channel tracking applications. Transitions between REs are defined by appropriate transition functions, which are chosen to be the same functions as for the transition to/from a visibility region, see below.

Generation of impulse responses

The structure of the COST 259-DCM, consisting of the cell type, the radio environment, and the random propagation scenarios, defines a principal framework from which models for the behavior of the MPCs can be deduced. This requires the definition of particular sets of local parameters and their statistics by means of global parameters. The latter definitions should be based on the propagation conditions encountered in a given radio environment.

Basically, the parameters of the DCIR can be given by means of a delay-angle distribution of the incident waves , or a scatterer geometry.

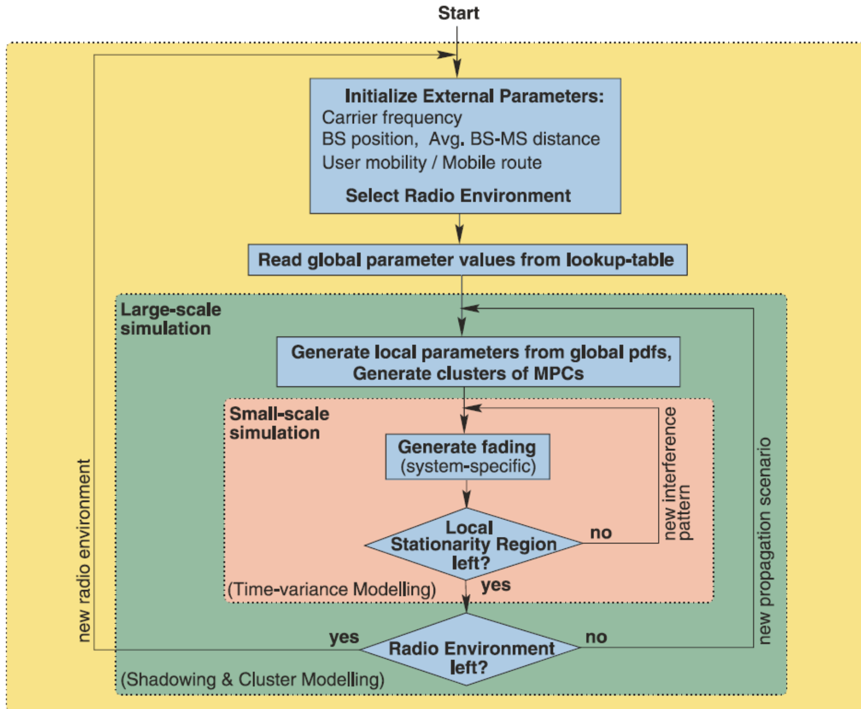

Figure 7.27: COST 259 flowchart. From [Molisch et al. 2006].

For the implementation based on the delay-angle distribution, we have to distinguish between the generation of local parameters and the creation of the complex impulse response. In a first step, we have to generate a small-scale PDDP (or equivalently, a parameter set), from large-scale information. This large-scale information can either be obtained from geometric considerations (see below), or the probability density function (PDF) of the number of impinging waves $f(L)$, the PDF of the multipath arrivals $f(\tau_\ell, \Omega_\ell)$, and the PDF of the power of the ℓ-th wave given their delay and incidence direction $f\left(|\alpha_\ell|^2 \,\middle|\, \tau_\ell, \Omega_\ell\right)$. Note that a *large-scale* PDDP

$$P_{\mathcal{R}}(\tau, \Omega) \;=\; \mathrm{E}_{A\in\mathcal{R}}\left\{\frac{1}{P_A} P_A(\tau, \Omega)\right\}, \tag{7.103}$$

where the expectation is taken over the local area A within the radio environment $\mathcal{R}$, gives insufficient information for the generation of the directional impulse responses with correct statistics. We need the statistics of the PDDPs, so that we can create proper realizations. Once we have such a realization, we can then generate complex CIRs, according to the methods described in Chapter 7. Specifically, we can either prescribe a cluster delay-angle profile and convert this into a generalized tapped delay line, or we can create a GSCM realization of the clusters and derived the CIR from that.

The COST 259 model - Macrocells

The MPCs in the macrocells occur in clusters. There always exists at least one cluster that corresponds to local wave interactions near the UE. The number of additional clusters is modeled as a Poisson distribution. The occurrence of additional clusters is determined by a number of visibility areas, circular areas with radius R_C , that are generated within the region of operation, see Sec. 7.6.4. Each time a UE enters into a visibility area, a corresponding cluster of MPCs is made active (the corresponding group of IOs becomes "visible"), and when the UE leaves the visibility area the cluster will be deactivated again. A smooth transition from non-active to active is achieved by scaling the relative power of the cluster by a factor A_m^2 . The transition function that is used is specified as

$$A_{\mathrm{m}}\left(\mathbf{r}_{\mathrm{UE}}\right) = \frac{1}{2} - \frac{1}{\pi}\arctan\left(\frac{2\sqrt{2}y}{\sqrt{\lambda x}}\right) \tag{7.104}$$

with

$$y = L_C + \left|\mathbf{r}_{\mathrm{UE}} - \mathbf{r}_{\mathrm{m}}\right| - R_C \tag{7.105}$$

and $x = L_C$. Here, $\mathbf{r}_{\mathrm{m}}q$ represents the center of the circular visibility area. The first cluster is always active. In order to give a constant expectation for the number of clusters that is equal to the associated value of N_C in Table 7.21, the area density of the visibility areas must be

$$\rho_C = \frac{N_C - 1}{\pi(R_C - L_C)^2} \quad \left[m^{-2}\right] \tag{7.106}$$

Each cluster is further described by its average azimuth and elevation (as seen from the BS) within a directional channel impulse response estimate and the delay of the first component in the cluster. In the model, an IO location, $\mathbf{r}_{C,m}$, is associated with each cluster. The location is arbitrarily drawn from a two-dimensional Gaussian distribution centered on the corresponding visibility area center, $\mathbf{r}_m$. The standard deviation of the distribution is equal to $|\mathbf{r}_m|$. The delay and azimuth of the cluster occurring due to reception of energy from a particular group of IOs is then given by applying a single-interaction geometrical construction

$$\phi_m = \begin{cases} \arg\left(\mathbf{r}_{\mathrm{UE}} - \mathbf{r}_{\mathrm{BS}}\right) & m = 1 \\ \arg\left(\mathbf{r}_{\mathrm{C},m} - \mathbf{r}_{\mathrm{BS}}\right) & m > 1 \end{cases} \tag{7.107}$$

$$\tau_m = \begin{cases} \frac{1}{c}\left|\mathbf{r}_{\mathrm{UE}} - \mathbf{r}_{\mathrm{BS}}\right| & m = 1 \\ \frac{1}{c_0}\left(\left|\mathbf{r}_{\mathrm{UE}} - \mathbf{r}_{\mathrm{C},m}\right| + \left|\mathbf{r}_{\mathrm{C},m} - \mathbf{r}_{\mathrm{BS}}\right|\right) & m > 1 \end{cases} \tag{7.108}$$

The description is described in greater detail in the COST 273 final report: In a first step, visibility regions for the clusters are distributed throughout the cell. The number of visibility regions in the cell is chosen in such a way that a randomly placed UE sees, on average, the mean number of additional clusters N_{C}. Each visibility region is associated with one specific cluster. Note that due to this approach, the total number of clusters in the cell can become quite large, but only a small part of them is visible (and thus needs to be taken into account for the computations) at each point in time. The position of the cluster relative to the position of the BS and the cluster center, is determined by the following geometric approach: draw a line from the BS to the center of the visibility region. The cluster position will be determined relative to that connection line. The radial distance from the BS is determined from an exponential distribution

$$f(r) = \begin{cases} 0 & r < r_{\min} \\ 1/(\sigma_r))exp(-((r - r_{\min})/(\sigma_r)) & \text{otherwise} \end{cases} \tag{7.109}$$

The angle of the cluster center is then drawn at random from a Gaussian distribution with a standard deviation $\sigma_{\phi,\mathrm{C}}$. This fixes the position of the cluster.

Motion of the UE, as reflected in changes in UE location, causes changes in the delay but not azimuth of the clusters. The exception is the first cluster, corresponding to local wave interactions only, the azimuth and delay of which will move with the UE.

The elevation angles to distant IOs are assumed to be zero. For the local cluster the elevation angle is calculated for NLOS conditions as

$$\theta_m = \begin{cases} \arctan\left(\frac{h_{\mathrm{B}} - h_{\mathrm{BS}}}{d}\right) & m = 1 \\ 0 & m > 1 \end{cases} \tag{7.110}$$

where the average building height h_{B} is an external parameter. For LOS, h_{B} is replaced with h_{UE}.

For the path gain, we first need the LOS probability. It is modeled as

$$p_{\mathrm{LOS}}(d) = \max\left(\frac{h_{\mathrm{BS}} - h_{\mathrm{B}}}{h_{\mathrm{BS}}} \frac{d_{\mathrm{co}} - d}{d_E}, 0\right) \tag{7.111}$$

where the cutoff distance d_{co} is an external parameter. To comply with the distance-dependent probability of line-of-sight, circular visibility regions with radius R_{L} are distributed with an area density ρ_{L}

$$\rho_{\mathrm{L}}(d) = \frac{p_{\mathrm{LOS}}(d)}{\pi (R_{\mathrm{L}} - L_{\mathrm{L}})^2} \quad \left[m^{-2}\right] \tag{7.112}$$

In the case of overlapping visibility areas the one giving the highest value of A_{L} is used.

Next, consider the power in different clusters. The model is formulated as

$$P_m = \begin{cases} S_1 \frac{P_{NLOS}}{s(d)} + A_L^2 \left(\frac{\lambda}{4\pi d}\right)^2 & m = 1 \\ A_m^2 S_m \frac{P_{NLOS}}{s(d)} 10^{-\frac{k_\tau \min(\tau_m - \tau_1, \tau_B)}{10}} & m > 1 \end{cases} \tag{7.113}$$

where P_{NLOS} is the NLOS path gain, A_{L} is the transition function for activating the LOS component, S_m is the cluster shadowing, and A_m the transition function for activating/de-activating a cluster. Parameters k_τ and τ_{B} describe the cluster path gain as a function of the delay between the first and the later clusters. The NLOS pathgain is obtained from the COST 231-Walfisch-Ikegami model for the urban environments GTU and GBU while the COST 231-Hata model is used for the rural environments GRA and GHT. The correction factor for Suburban Areas should be used in the COST 231-Hata model.

The DDADPS of each cluster is modeled as

$$P(\tau, \theta, \phi, \theta', \phi') = P_\tau(\tau) P_\theta(\theta) P_\phi(\phi) P_{\theta'}(\theta', \tau) P_{\phi'}(\phi', \tau) \tag{7.114}$$

Here, the PDP is modeled as single-exponential, the azimuth and elevation spectra at the BS as Laplacian (with different spreads), while at the UE, the azimuth distribution depends on the delay

$$P_{\phi'}(\phi', \tau) = \begin{cases} \frac{1}{2\pi} & \tau < \tau_1 + \tau_C \\ \frac{1}{\sigma_{\phi'} 2\sqrt{2}} \left(e^{-\frac{\sqrt{2}|\phi' - \phi'_A|}{\sigma_{\phi'}}} + e^{-\frac{\sqrt{2}|\phi' - \phi'_B|}{\sigma_{\phi'}}} \right) & \tau \geq \tau_1 + \tau_C \end{cases} \tag{7.115}$$

where $\sigma_{\phi'}$ is the azimuth spread and ϕ'_A and ϕ'_B are the directions of the street canyon in which the UE is located. These directions can be obtained directly from road orientation ϕ_R. Delayed clusters are characterized by a single Laplacian distribution around the direction ϕ'_m of the associated IO:

$$P_{\phi'}(\phi', \tau) = \frac{1}{\sigma_{\phi'}\sqrt{2}} e^{-\sqrt{2}|\phi' - \phi'_m|/\sigma_{\phi'}} \tag{7.116}$$

Elevation angles are modeled using a uniform distribution between 0 and a delay-dependent maximum angle

$$P_{\theta'}(\theta', \tau) = \frac{1}{\theta'_{\max}(\tau)} \quad 0 \leq \theta' \leq \theta'_{\max}(\tau) \tag{7.117}$$

For rural environments (GRA,GHT) the maximum elevation angle is a constant. For urban environments (GTU,GBU) it is

$$\theta'_{\max}(\tau) = \frac{1}{1 + (\tau - \tau_1)/\tau_{\theta'}} \arctan\left(\frac{2h_B}{w}\right) \tag{7.118}$$

For Rician fading, a single component at the delay and angle of the geometrical LOS is added.

Parameter	Symbol	Valid range	Typical value
Carrier frequency [Hz]	f	150 MHz-2 GHz (GRA,GHT) 800 MHz-2 GHz (GTU,GBU)	
Bandwidth [Hz]	BW	0-5 MHz	
Base station height [m]	h_{BS}	30-200 m (GRA,GHT) 4-50 m (GTU,GBU)	50 m (GBU,GRA,GHT) 30 m (GTU)
Mobile station height [m]	h_{MS}	1-10 m (GRA,GHT) 1-3 m (GTU,GBU)	1.5 m
Base station position [m]	$\overline{r}_{BS}$	Any	
Mobile station position [m]	$\overline{r}_{MS}$ $d = \lvert\overline{r}_{MS} - \overline{r}_{BS}\rvert$	$d < 20$ km (GRA,GHT) $d < 5$ km (GTU,GBU)	$d = 5000$ m (GRA,GHT) $d = 500$ m (GTU,GBU)
Average building height [m]	h_B	8-60 m, $h_B < h_{BS}$	15 m (GTU), 30 m (GBU)
Width of roads [m]	w	Any	$b/2$
Building separation [m]	b	Any	30 m (GTU), 50 m (GBU)
Road orientation with respect to direct path	φ_R	0-90°	45°

Table 7.20 External parameters of the COST 259 model.

Environment	Fraction of total meas. time with			Average # of clusters N_C
	1 cluster	2 clusters	3 clusters	
Bad Urban	0.27	0.28	0.45	2.18
Typical Urban	0.87	0.09	0.04	1.17
Suburban Area	0.92	0.08	0.00	1.08
Rural Area	0.94	0.06	0.00	1.06

Table 7.21 Number of clusters.

The cluster shadowing, angular spread, and delay spread are modeled similar to the 3GPP model (actually, the COST 259 model came first, and 3GPP adopted the approach):

$$\sigma_{\phi,m} = \quad m_{s\phi} 10^{s_{s\phi} Y_m/10} \tag{7.119}$$

$$\sigma_{\phi,m} = \quad m_{s\phi} 10^{s_{s\phi} Y_m/10} \tag{7.120}$$

$$\sigma_{\tau,m} = \quad m_{s\tau} \left(\tfrac{d}{1000}\right)^{\varepsilon} 10^{s_{s\tau} Z_m/10} \tag{7.121}$$

where X_m, Y_m, and Z_m are random Gaussian variables with zero mean, unit variance, and cross-correlations $\rho_{XY}, \rho_{XZ}, \rho_{YZ}$, and each of the random variables have exponential autocorrelation functions with autocorrelation lengths $\{L_S, L_\tau, L_\phi\}$. The BS elevation spread is assumed to be independent of all other cluster spreads.

For the small-scale fading, the statistics are assumed to be Rician, characterized by the narrow-band Rice factor K_0, which is described as a lognormal parameter with mean

$$m_K = \frac{26 - L_E\,[dB]}{6} \tag{7.122}$$

where L_E is the excess path loss (beyond the free-space pathloss). The variance (in dB) is 6, and the correlation length 8 m.

Microcells

The microcell model in COST 259 is based on the concept of Virtual Cell Deployment Areas (VCDAs). Essentially, a VCDA is a map of a synthetic city region. It contains street canyons,

	GTU / GBU	GRA / GHT
R_C [m]	100	300
L_C [m]	20	20

Table 7.22. Cluster radius and transition region.

		GTU / GBU	GRA / GHT
NLOS path gain	P_{NLOS}	From COST 231-Walfisch-Ikegami [22]	From COST 231-Hata "Suburban" [22]
Correction factor (determined numerically)	$s(d)$	GTU: $1.36 \cdot d^{-0.03}$ GBU: $5.1 \cdot d^{-0.15}$	GRA: $1.07 \cdot d^{-0.004}$ GHT: $5.1 \cdot d^{-0.15}$
Cluster power	k_τ [dB/μs]	1	1
	τ_B [μs]	10	10
LOS occurrence	d_{co} [m]	500	5000
	R_L [m]	30	100
	L_L [m]	20	20

Table 7.23 . Path gain parameters

	GTU / GBU	GRA / GHT
$\sigma_{\varphi'}$ [°]	10	10
τ_C [μs]	0.4	∞
θ'_{max} [°]	From eq. (22)	15
$\tau_{\theta'}$ [μs]	3	3

Table 7.24 . DoA parameters at the UE

Parameter (source for value)		GTU / GBU	GRA / GHT
Shadow fading (Table VIII)	s_{shf} [dB]	6	6
Azimuth spread (Table VI)	$m_{s\varphi}$ [°] $s_{s\varphi}$ [dB]	10 3	5 3*
Elevation spread (Table VII)	$m_{s\theta}$ [°] $s_{s\theta}$ [dB]	0.5 3*	0.25 3*
Delay spread (Greenstein [29])	$m_{s\tau}$ [μs] ϵ $s_{s\tau}$ [dB]	0.4 0.5 3	0.1 0.5 3
Autocorrelation distances (Table VIII,[58])	$L_S, L_\tau, L\varphi, L_\varphi$ [m]	100(*)	100(*)
Cross-correlations ([29],[58])	ρ_{XY} ρ_{XZ} ρ_{YZ}	-0.75 -0.75 0.5	-0.75* -0.75* 0.5*

Table 7.25 . Cluster spread parameters

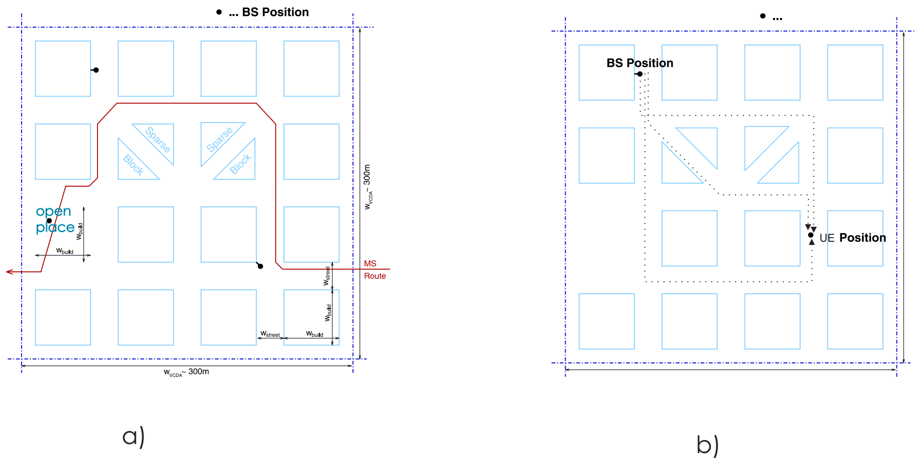

Figure 7.28: Typical example of a VCDA with a UE trajectory (left), and paths of clusters from the BS to the UE.

street crossings, open places, and far scatterers. Depending on the specifications and the location of the BS and UE, any of the four microcell environments (street canyon, street crossing, open place, open place NLOS) might be "active". Figure 7.28a shows the VCDA. In addition to the map, we also specify a route of the UE through this city region.

At each location of the UE, the signal can reach the UE through different street canyons, see Figure 7.28b. Each of these gives rise to a cluster; the inter-cluster characteristics are thus determined by the geometry, very similar to the semi-deterministic channel models described in Sec. 7.7 and used in the METIS and MiWeBa models (again, COST 259 came first; the idea was later adopted by METIS and MiWeBa). In the COST 259 models, the cluster parameters are chosen to be the same as for the TU macrocell model, though pathloss is chosen somewhat differently. Since the COST259 microcell model is in less use than the macrocell, we refrain from listing a complete parameterization and refer to the COST 259 final report [Correia 1999] for details.

Modifications in the COST 273 and COST 2100 Models

The main modification in the COST 273 model was the introduction of the twin clusters, which allowed an independent setting of the cluster DoAs and DoDs even with the geometrical approach, see Sec. 7.6.3. Furthermore, the concept of DMC (see Sec. 7.5.4) was incorporated into the model. However, a parameterization was restricted to urban macro (and urban micro) cells, and also for this case most parameters were taken from the COST 259 model, see [Molisch and Hofstetter 2006].

The main addition in the COST 2100 model is the multi-link model as described in Sec. 7.6.5. Parameterization for indoor and outdoor (without further differentiation) are given in Sec. 3.6 of the COST 2100 final report [Verdone and Zanella 2012], though it must be noted that the outdoor parameterization is based on a peer-to-peer measurement campaign at 300 MHz, and thus the parameters might not be representative of the more typical cellular scenarios. For completeness, Tables 7.26-7.31 provide the parameterization for the COST 273 and 2100 model. Most of the parameters are known from the discussion above; additional values are described in the table captions.

Further conceptual extensions have been done in the COST actions IC 1004, where a model was introduced that extends the visibility region to the BS, such that, e.g., in a massive MIMO array only part of the array might lie in in the visibility region, and the implementation of adding DMC was discussed. COST IRACON presented a large number of measurements and derived models in a variety of environments, see [Oestges and Quitin 2021], but did not provide explicit expansions

Parameter	Unit	Values					
		Indoor			Outdoor		
		LOS [PHL+10b]	NLOS [PHL+10b]	NLOS [QOHD09c]	LOS [ZTW+10]	NLOS [ZTW+10]	COST273 macrocell [Cor06b]
f_c	GHz	5.3	5.3	3.6	0.285	0.285	0.9 to 2.0
B	MHz	60	60	200	20	20	
R_{ss}	s	3.9×10^{-2}	3.9×10^{-2}		1.4×10^{-3}	1.4×10^{-3}	
N_{ss}		1500	300	16	332	113	
$\mathbf{v}_{MS}$	m/s	$(-2.6, 0, 0)$	$(0.1, 0, 0)$		$(-0.2, 0.9, 0)$	$(1.0, 0, 0)$	
[3] h_{BS}	m	2.0	2.0	1.6	1.8	1.8	50
h_{MS}	m	1.4	1.4	1.5	2.1	2.1	1.5
[4] $\mathbf{r}_{MS}$	m	$(5, -5)$	$(24, 11)$		$(267, -222)$	$(78, 314)$	Random drop
$h_{rooftop}$	m				15	15	15
w_r	m				7	7	25
w_b	m				250 to 300	250 to 300	50
ϕ_{road}	°				30 to 90	30 to 90	45
n_{floor}		0	0	0			
r_{cell}	m	50	50	50	1000	1000	1000
$\mathbf{r}_{BS}$	m	$(0, 0, h_{BS})$ in all cases					

Table 7.26 External parameters for the COST 273 and COST 2100 model. R_{ss}: rate at which channel snapshots are taken, N_{ss} number of snapshots in the simulation, r_{MS} starting position of the UE, w_b width between buildings, w_r road width, ϕ_{road} orientation of road w.r.t LOS. From [Verdone and Zanella 2012].

Parameter	Unit	Values					
		Indoor			Outdoor		
		LOS [PHL+10b]	NLOS [PHL+10b]	NLOS [QOHD09c]	LOS [ZTW+10]	NLOS [ZTW+10]	COST273 macrocell [Cor06b]
$\lambda_{N_{C,add}}$		0.8	0.7	1.7			
N_{const}		3.7	4.3	3.0	4.0	4.0	1.2
R_C	m	2.8	3.8		6.3	5.2	100
L_C	m	0.3	0.4		1.4	1.9	20
k_τ	dB/μs	54.2	5.9	[7] > 12	16.9	3.2	1
τ_B	μs	1	1				10
K_{sel}		0.7	0.0		0.4	0.3	1
d_{co}	m	100					500
R_L	m	10					30
L_L	m	1					20
μ_K	dB	0.4	-9.7		-31.5	-27.0	[8] $(26 - EPL)/6$
σ_K	dB	2.8	7.4		8.2	8.7	6
σ_S	dB	4.9	5.4	5.1	0.4	0.5	6

Table 7.27 Stochastic parameters for the COST 273 and COST 2100 model. All definitions like in COST 259. From [Verdone and Zanella 2012].

Parameter	Unit	Values	
		Indoor	
		LOS [PHL+10b]	NLOS [PHL+10b]
$\mu_{\tau_{C,link}}$	ns	44.1	74.8
$\tau_{C,link,min}$	ns	27.3	14.0
$\mu_{d_{C,BS}}$	m	4.2	8.5
$d_{C,BS,min}$	m	6.7	8.0
$\mu_{d_{C,MS}}$	m	8.4	6.7
$d_{C,MS,min}$	m	1.5	1.1
$\mu_{\Psi_{C,AZ,BS}}$	°	−17.1	28.4
$\sigma_{\Psi_{C,AZ,BS}}$	°	47.0	30.6
$\mu_{\Psi_{C,AZ,MS}}$	°	−7.6	−30.2
$\sigma_{\Psi_{C,AZ,MS}}$	°	38.8	52.0
$\mu_{\Psi_{C,EL,BS}}$	°	1.2	−4.8
$\sigma_{\Psi_{C,EL,BS}}$	°	23.5	6.4
$\mu_{\Psi_{C,EL,MS}}$	°	20.3	−0.1
$\sigma_{\Psi_{C,EL,MS}}$	°	38.2	10.2

Table 7.28 Cluster location parameter for the COST 2100 model. $\tau_{\mathrm{C,link}}$ is the extra delay for twin clusters; it is single-exponentially distributed with minimum delay $\tau_{\mathrm{C,link,min}}$ and mean $\mu_{\tau_{\mathrm{C,link}}}$. Cluster angle ψ defines directions of clusters seen from the BS and UE, respectively. Gaussian distribution with mean μ_ψ and standard deviation σ_ψ. From [Verdone and Zanella 2012].

Parameter	Unit	Values					
		Indoor			Outdoor		
		LOS [PHL+10b]	NLOS [PHL+10b]	NLOS [QOHD09c]	LOS [ZTW+10]	NLOS [ZTW+10]	COST273 macrocell [Cor06b]
$\mu_{N_{\mathrm{MPC}}}$		4	4	10	4	4	20
$m_{\psi_{\mathrm{BS}}}$	°	0.55	2.8		17.1	19.8	6.5
$S_{\psi_{\mathrm{BS}}}$	dB	3.1	3.6		2.0	2.0	0.34
$m_{\theta_{\mathrm{BS}}}$	°	1.4	2.6				3.2
$S_{\theta_{\mathrm{BS}}}$	dB	3.4	3.8				3
$m_{\psi_{\mathrm{MS}}}$	°	2.8	6.0	8.7	17.8	21.3	35
$S_{\psi_{\mathrm{MS}}}$	dB	3.6	1.9	1.1	2.2	1.8	0
$m_{\theta_{\mathrm{MS}}}$	°	3.2	1.3	7.5			$\sqcap(0,45)$
$S_{\theta_{\mathrm{MS}}}$	dB	2.3	2.5	0.84			0
m_{τ}	ns	0.81	4.2	3.7	135	247	400
S_{τ}	dB	3.3	1.4	1.8	2.5	2.5	3
$\rho_{\psi_{\mathrm{BS}}\theta_{\mathrm{BS}}}$		0.6	0.6				
$\rho_{\psi_{\mathrm{BS}}\psi_{\mathrm{MS}}}$		0.6	−0.2		0.3	0.4	0.0
$\rho_{\psi_{\mathrm{BS}}\theta_{\mathrm{MS}}}$		0.6	0.3				
$\rho_{\psi_{\mathrm{BS}}\tau}$		0.6	−0.2		0.6	0.6	0.5
$\rho_{\psi_{\mathrm{BS}}\sigma_{\mathrm{S}}}$		0.1	0.0		0.0	0.0	−0.6
$\rho_{\theta_{\mathrm{BS}}\psi_{\mathrm{MS}}}$		0.3	−0.2				
$\rho_{\theta_{\mathrm{BS}}\theta_{\mathrm{MS}}}$		0.7	0.0				
$\rho_{\theta_{\mathrm{BS}}\tau}$		0.6	−0.2				
$\rho_{\theta_{\mathrm{BS}}\sigma_{\mathrm{S}}}$		−0.1	0.3				
$\rho_{\psi_{\mathrm{MS}}\theta_{\mathrm{MS}}}$		0.5	0.2				
$\rho_{\psi_{\mathrm{MS}}\tau}$		0.6	0.8		0.8	0.9	0.0
$\rho_{\psi_{\mathrm{MS}}\sigma_{\mathrm{S}}}$		0.0	0.1		0.1	0.0	0.0
$\rho_{\theta_{\mathrm{MS}}\tau}$		0.5	0.5				
$\rho_{\theta_{\mathrm{MS}}\sigma_{\mathrm{S}}}$		0.0	0.0				
$\rho_{\tau\sigma_{\mathrm{S}}}$		−0.1	0.2		0.0	−0.2	−0.6

Table 7.29 Cluster spread parameters for the COST 273 and COST 2100 model. N_{MPC} is number of MPCs in a cluster. From [Verdone and Zanella 2012].

Parameter	Unit	Values			
		Indoor			
		LOS [PHL+10b]	NLOS [PHL+10b]	NLOS [QOHD09c]	COST273 macrocell [Cor06b]
$m_{\mu_{\mathrm{CPR}}}$	dB	0.9	−5.3	4.2	0
$S_{\mu_{\mathrm{CPR}}}$	dB			8.2	
$m_{\sigma_{\mathrm{CPR}}}$	dB	6.6	11.9	$1.3+\mu_{\mathrm{CPR}}$	$-\infty$
$S_{\sigma_{\mathrm{CPR}}}$	dB			3.3	
$m_{\mu_{\mathrm{XPD_V}}}$	dB	15.6	9.6	13.5	6
$S_{\mu_{\mathrm{XPD_V}}}$	dB			7.8	
$m_{\sigma_{\mathrm{XPD_V}}}$	dB	10.4	9.5	7.8	2
$S_{\sigma_{\mathrm{XPD_V}}}$	dB			$1.3+\mu_{\mathrm{XPD_V}}$	
$m_{\mu_{\mathrm{XPD_H}}}$	dB	14.9	11.4	3.9	6
$S_{\mu_{\mathrm{XPD_H}}}$	dB			5.5	
$m_{\sigma_{\mathrm{XPD_H}}}$	dB	11.8	8.7	$1.3+\mu_{\mathrm{XPD_H}}$	2
$S_{\sigma_{\mathrm{XPD_H}}}$	dB			3.3	

Table 7.30 Polarization parameters for the COST 2100 model. From [Verdone and Zanella 2012].

Parameter	Unit	Values	
		Indoor	
		LOS [PHL+10b]	NLOS [PHL+10b]
$p_{2,\mathrm{BS}}$		0.02	0.56
N_{VR}		2.0	1.4

Table 7.31 Multi-cluster parameters for the COST 2100 model. $p_{m,BS}$ is probability for an m-BS common cluster; N_{VR} is the mean of the (Poisson-distributed) number of VRs in a VR group (defined such that all UEs in VRs that make up a VR group see the same clusters. From [Verdone and Zanella 2012].

of the COST 259/273/2100 model.

References

Correia 1999 L. M. Correia (ed), "Wireless flexible personalized communications", Final report of the COST 259 action, John Wiley & Sons, Inc. (2001).
Correia 2006 L. M. Correia (ed), "Mobile broadband multimedia networks: techniques, models and tools for 4G", final report of the COST 273 action, Elsevier (2006).
Verdone and Zanella 2012 R. Verdone and A. Zanella (eds), "Pervasive Mobile and Ambient Wireless Communications", Final report of the COST 2100 action, Springer (2012).
Cardona 2016 N. Cardona (ed), "Cooperative Radio Communications for Green Smart Environments", Final report of the COST IC1004 action, River Publishers (2016).
Oestges and Quitin 2021 C. Oestges and F. Quitin, "Inclusive Radio Communications for 5G and Beyond", Final report of COST IRACON, Academic Press (2021).